

BizRadar v4

Universal Sector Algorithm

Perfect-fit specifications for every NAICS sector — built so a single platform can analyze any of 1,464 business types with the same rigor it applies to a hotel.

1,464 business types	20 NAICS sectors (18 in scope)	63 sub-cluster specs	5 implementation phases
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Why this document exists

BizRadar v3 is a hospitality intelligence engine pretending to be universal. It works beautifully for hotels, restaurants, salons, gyms — and falls apart for the other ~67% of businesses on the master list. v4 introduces a Master Router (Layer 0.5) that classifies every input by business model and routes it to a sector-specific pipeline. The pipeline architecture stays the same; the contents of every layer are swapped per sector. This document is the complete specification of those contents — one chapter per sector, plus the universal scaffolding and the implementation phases.

How to read this

Part 1 is the architecture and the router — read top to bottom. Part 2 has one chapter per sector — read the chapters relevant to your phase. Part 3 is the failure-rate database. Part 4 is the implementation roadmap with explicit phase gates. Part 5 is reference and appendices. Engineers should pair this document with the existing v3 server.js — every section maps to a code module that needs to be added or modified.

Promise of this spec

Every sector chapter is structured identically: business-model classification, sub-clusters, data sources, review taxonomy, opportunity playbook, specificity rules, cost scale, persona schema, cycle model, issue library, report sections, KPIs, and out-of-scope notes. If a section says 'not applicable' for a sector, that is a deliberate engineering decision — not an oversight. The router is responsible for routing requests to the correct chapter; the chapter is responsible for producing a useful report.

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PART 1 — UNIVERSAL ARCHITECTURE

1.1 The flaw of v3 in one diagram

BizRadar v3 has one pipeline. Every business — a hotel, a CPA firm, a steel forge, a SaaS company, a hog farm — runs through the same nine layers with the same hospitality-shaped contents inside. The architecture is sound; the contents are wrong for everything except hospitality.

L0 Input	Classify input type	Business name regex includes hotel chains	Trucking firm or CPA firm gets misrouted
L1 Data	9 parallel APIs	Google + TripAdvisor + Foursquare	B2B firms have ~0 reviews on these
L2A Reviews	Synthesize reviews	Hotels have hundreds of reviews	Manufacturer has 3 — analysis is noise
L2A Categories	Tag review categories	food, ambiance, cleanliness, WiFi	Wrong taxonomy for a dental practice
L2A Competitors	Top 3 within 2.5km	Walk-in radius	B2B competes statewide or nationally
L3 Opportunities	18 categories of ideas	Sensory experience, photo moments, scents	Meaningless for a wholesaler or a SaaS
L4 Hyper-local	Named producers within 60mi	Carr Valley Cheese style local	B2B 'local' is industries, not producers
L5 Scoring	Cost scale 1–5 = \$0 to \$10k	Hotel-scale spending	Manufacturer evaluates in \$100k–\$10M
L6 Personas	Sarah, 45, parent for graduation	B2C consumer schema	B2B buyer has committee, not lifestyle
L7 Seasonal	Calendar quarters	Tourist seasons	Construction, B2B run on different cycles
L8 Issues	WiFi, bedbugs, noise	Hospitality complaint library	HVAC complaints are no-shows + surprise pricing
L9 Output	9 fixed sections	Hospitality report layout	B2B needs RFP pipeline + cert roadmap

The fix in one sentence

Keep the 9-layer architecture. Insert a Master Router at Layer 0.5 that selects which **contents** populate each subsequent layer based on the business model class. The architecture is universal. The contents are sector-specific. This document specifies both.

1.2 The v4 architecture — router plus 9 layers

v4 inserts Layer 0.5 — the Master Router — between input classification (L0) and data collection (L1). The router reads the business model class from L0, then selects the appropriate sector profile. Every downstream layer then loads its contents from the selected profile.

```

USER INPUT
  |
  v
L0  Input Classifier -> detect: location | city | niche | url | ambiguous
  |
  v
L0.5 MASTER ROUTER (NEW IN v4)
    reads: name + Google Places category + NAICS keyword + Part2 lookup
    outputs: { naics_code, business_model, sector_profile_id }
    loads:  sector_profiles[profile_id] -> used by every layer below
  |
  v
L1  Parallel Data Collection
    APIs and competitor radius come from the loaded profile
    e.g. hotel = Google + TA + Yelp / 2.5km
         HVAC  = Google + Angi + BBB / 30 mi
         SaaS  = G2 + Capterra + ProductHunt / global
  |
  v
L2A Reviews & Competitor Analysis
    review categories come from profile
    commonality detection unchanged
  |
  v
L3  Opportunity Engine
    18 categories <-- HOSPITALITY playbook
    12-20 categories <-- per-sector playbook (one of 18 sets)
  |
  v
L4  Specificity Layer (was 'Hyper-Local' in v3)
    what counts as a real named entity comes from profile
  |
  v
L5  Scoring & Ranking
    cost scale relative to revenue band, set by profile
  |
  v
L6  Persona Engine
    schema picked by profile: B2C-consumer | B2B-buyer | patient | client
  |
  v
L7  Cycle Model (was 'Seasonal' in v3)

```

```

    cycle definition picked by profile
    |
    v
L8  Issue Library
    per-sector category-standard complaints loaded from profile
    |
    v
L9  Output Delivery
    section list picked by profile (some appear / disappear by sector)
    |
    v
REPORT DELIVERED TO USER

```

What this means in code terms

v4 introduces a new file: **sector_profiles.js**. It exports a dictionary keyed by **sector_profile_id**. Each value is a complete profile object with fields for: data sources, review categories, opportunity playbook, specificity rules, cost scale, persona schema, cycle model, issue library, and report sections. The Master Router selects one profile per request. `server.js` is refactored to read from the loaded profile rather than from hardcoded hospitality logic.

1.3 Layer 0.5 — Master Router (full spec)

The Master Router runs after L0 and before L1. It outputs three values that determine the entire downstream pipeline: **naics_code** (6-digit), **business_model** (one of nine classes), and **sector_profile_id** (the lookup key into sector_profiles).

Inputs to the router

- **Business name** — from L0 (e.g. 'Smith Plumbing', 'AmericInn Dodgeville')
- **Google Places category** — from L1 once it returns (router can run after L1 for ambiguous cases)
- **NAICS keyword library** — appendix E in this document, mapping ~3,000 keywords to NAICS codes
- **Part 2 master list** — 452 modern business types that map to NAICS parents
- **User override (optional)** — explicit NAICS code if user is sophisticated

Classification rules — order matters

- 1 Try exact match against Part 2 master list (e.g. 'food truck' -> 722330, 'yoga studio' -> 713940).
- 2 Try Google Places primary category mapping (e.g. 'lodging' -> 721110, 'plumber' -> 238220).
- 3 Try keyword regex from Appendix E (e.g. 'CPA' / 'tax' / 'accountant' -> 541211).
- 4 If still ambiguous, surface a confirmation question to the user before any L1 calls fire.
- 5 Record confidence score for the classification — under 0.6 triggers a soft-confirm dialog.

The nine business model classes

B2C_walk_in	Customer physically walks in, mostly without appointment	Foot traffic, on-premise consumption	722511, 445110, 812111
B2C_appointment	Consumer appointments, sometimes drop-in	Scheduling system, professional license	621210, 541110, 812112
B2C_online	Direct-to-consumer online only	No physical address that matters	454110, 519290, 513210
B2B_local	Sells to local businesses, geography matters	Service area, route density	238220, 561720, 484110
B2B_regional	Sells to businesses in a multi-state region	Distributor / wholesale model	423110, 484121, 561311
B2B_national	National or global B2B sales	Trade show circuit, named OEMs	33xxxx, 5415xx, 423430
B2G	Sells primarily to government entities	SAM.gov registration, set-aside certs	237310, 541330, 561612
Regulated_monopoly	Geographic monopoly under regulator	PUC / FERC oversight	221xxx, 4861xx, 4862xx

Out_of_scope	Public administration, holding companies	Not a customer-facing business	5511xx, 9211xx-9281xx

Sector profile selection

Once class is determined, the router picks one of 60+ sector profile IDs. The profile ID is more granular than the NAICS sector — many sectors have multiple profiles (e.g. construction has separate profiles for residential trades, commercial GCs, and civil/heavy). The full mapping is in Appendix A.

Pseudocode

```
function classify(input) {
  // Step 1: extract candidate NAICS
  let candidates = [];
  candidates.push(...part2_lookup(input.name));          // food truck -> 722330
  candidates.push(...gplaces_to_naics(input.gcategory)); // 'plumber' -> 238220
  candidates.push(...keyword_regex(input.name));         // 'CPA' -> 541211
  if (candidates.length === 0) return ASK_USER;

  // Step 2: pick highest-confidence candidate
  let { naics, confidence } = best(candidates);
  if (confidence < 0.60) return SOFT_CONFIRM(naics);

  // Step 3: derive business model + profile
  let business_model = naics_to_model[naics.slice(0,4)];
  let profile_id     = naics_to_profile[naics.slice(0,4)];

  // Step 4: return full classification
  return { naics, business_model, profile_id, confidence };
}
```

Failure modes the router must handle

Ambiguous name	'Smith & Sons' could be 50 things	Soft-confirm dialog — show top 3 NAICS guesses
Out-of-scope sector	User pastes a government office	Friendly rejection with explanation, refund flow
Multi-line-of-business	Hotel with restaurant inside	Pick primary NAICS, note secondary in report
Brand new format	AI-driven business with no NAICS yet	Map to closest parent + add to Part 2 backlog
User override mismatch	User claims NAICS that conflicts with Google	Use user override but flag the mismatch in QA log

1.4 Universal scaffolding — what stays constant across all sectors

These elements do not change between profiles. They are the bones that hold the system together. Anything not on this list *must* be sector-customized.

Constants across every sector

Layer execution order	L0 -> L0.5 -> L1 -> L2A -> L3 -> L4 -> L5 -> L6 -> L7 -> L8 -> L9	Section 1.2
Promise.all() data collection	Parallelism is the right pattern regardless of which APIs are called	L1
8-second hard timeout per source	Fail fast principle is universal	L1
Recency weighting on review scores	More recent feedback is more relevant in every sector	L2A
KNOWN vs HIDDEN issue classification	Known to all competitors -> capped. Unique -> high priority	L8
Final scoring formula shape	$(\text{Impact} \times 2 + \text{Novelty} \times 1.5 + \text{Mention} \times 1) / (\text{Ease} + \text{Cost})$	L5
Quick Win definition (relative)	High impact + low cost + low ease — content-relative thresholds	L5
Specificity rule (zero-tolerance)	Every recommendation names a real entity — what kind varies	L4
Confidence score on output	Reports always include data quality + recommendation confidence	L9
Cache 24h on identical input	Cost / latency optimization is universal	L1

Quality gates that fire for every report

- **Specificity gate:** if any recommendation contains a banned generic phrase ('local farm', 'nearby business', 'community partner'), force re-generation.
- **Volume gate:** minimum 12 ranked recommendations, minimum 3 personas, minimum 3 cycle entries — pad does not count.
- **Evidence gate:** every issue must have at least one quote (or one structured datum for B2B) under 15 words.
- **Confidence gate:** if data quality average is below 0.4, prepend a yellow warning to the report.
- **Out-of-scope gate:** if business_model is 'Out_of_scope', do not generate — return refund-eligible message.
- **Hallucination gate:** every named entity (producer, partner, certification) must be verifiable in a structured source — not free-text Claude output.

What "sector-perfect" means operationally

A report is sector-perfect when (a) every layer's contents come from the right profile, (b) every quality gate passes, (c) the user reads three recommendations and says "I haven't heard any of this before — and it's true". Generic, hospitality-shaped, or wrong-radius output fails this test and ships zero value to a non-hospitality customer.

1.5 The substitution principle — what changes per sector

Each sector profile is a JSON-serializable object with the following ten fields. Every chapter in Part 2 fills in these ten fields for one or more sub-clusters of a sector.

1	data_sources	list[Source]	Which APIs L1 fires, and the fallback chain
2	competitor_radius	string mi rule	How L1 defines 'nearby competitor'
3	review_categories	list[string]	What L2A tags reviews with
4	opportunity_playbook	list[Category]	The 12-20 categories L3 researches
5	specificity_rules	list[Rule]	What counts as a real named entity in L4
6	cost_scale	ScaleSpec	Currency bands for L5 'Cost' dimension
7	persona_schema	SchemaName	Field set for L6 personas
8	cycle_model	CycleSpec	L7 cycle definition (calendar / fiscal / event-driven)
9	issue_library	list[KnownIssue]	L8 category-standard complaints with quote templates
10	report_sections	list[SectionId]	L9 section list, in the order they render

Profile object shape (TypeScript-ish)

```
interface SectorProfile {
  id: string; // e.g. 'hospitality.hotel'
  display_name: string; // 'Hotels & Motels'
  naics_codes_covered: string[]; // ['721110', '721120', '721191', ...]
  business_model: BusinessModel;
  data_sources: Source[]; // ordered, with fallback chain
  competitor_radius: RadiusRule; // miles | service_area | national
  review_categories: string[]; // 6-12 tags for L2A
  opportunity_playbook: Category[]; // 12-20 entries
  specificity_rules: Rule[]; // what L4 must name
  cost_scale: { unit: 'USD', bands: [number, number, number, number] };
  persona_schema: 'B2C' | 'B2B' | 'patient' | 'client';
  cycle_model: { kind: 'calendar' | 'fiscal' | 'event'; entries: Cycle[] };
  issue_library: KnownIssue[];
  report_sections: SectionId[]; // ordered list of L9 sections
  out_of_scope_naics: string[]; // 6-digit NAICS to refuse even if matched
}
```

Every sector chapter in Part 2 is essentially a verbose, human-readable filling-in of this object. An engineer can read a chapter and translate it directly to the JSON profile.

1.6 Quality gates — universal pre-flight checks

These gates run after L9 builds the report but before it is shown to the user. Any gate that fails either triggers re-generation (soft fail) or blocks the report and refunds the user (hard fail).

Specificity	Zero generic phrases (banned word list)	Soft	Re-prompt L3 / L4 with stricter instructions
Volume	Min 12 recs, 3 personas, 3 cycle entries	Soft	Re-prompt with minimum count enforcement
Evidence	Every issue has a quote / data source	Soft	Drop unsupported issues, re-prompt for replacements
Confidence	Avg data quality > 0.4	Hard	Prepend warning banner; user can still view
Out of scope	business_model not in {Out_of_scope}	Hard	Refund + explain; do not generate
Hallucination	Every named entity is in structured source	Hard	Strip unverified entities, re-prompt to fill gaps
Coverage	All 10 profile fields populated	Hard	Engineering alert — profile is incomplete
Cost sanity	No recommendation outside cost band	Soft	Re-rank with cost band enforced
Persona match	All personas use the profile's schema	Soft	Re-prompt L6 with schema reference
Section presence	All required sections non-empty	Soft	Either fill or remove from report

Why hallucination is a hard fail

If BizRadar tells a Wisconsin hotel to partner with 'Driftless Cheese Co.' and that company does not exist, the user catches it on the first phone call and never trusts the product again. v3 has caught this exact failure mode three times in testing. v4 enforces structured-source verification: every named entity must be present in Google Places, Census Business Patterns, USDA producer registry, ThomasNet, or another verifiable source. Free-text Claude output without source linkage is rejected.

BUILD STATUS — BATCH 2 OF 5

What is in this version

This is Batch 2 of the BizRadar v4 spec. Front matter (Part 1) is complete. Eight of the eighteen in-scope sector chapters are now complete — the entire Phase 0 / Phase 1 / Phase 5 specialty roster. The remaining ten sectors are queued and will be appended in Batches 3-5. Parts 3 (failure-rate database), 4 (implementation roadmap), and 5 (appendices) are queued for the final batch.

What is complete (Batches 1-2)

Part 1 — Universal Architecture	Done — 6 sections covering router, scaffolding, substitution principle, quality gates
Ch. 11 — Agriculture, Forestry, Fishing	Done (Batch 1) — 3 sub-clusters: commodity / agritourism / forestry
Ch. 21 — Mining, Oil & Gas	Done (Batch 1) — 2 sub-clusters: aggregates / oil-gas-metals
Ch. 22 — Utilities	Done (Batch 1) — 1 sub-cluster: electric/gas/water utility
Ch. 72 — Accommodation & Food Services	Done (Batch 2) — 3 sub-clusters: lodging / restaurants / bars-breweries
Ch. 44-45 — Retail Trade	Done (Batch 2) — 3 sub-clusters: brick-and-mortar / auto dealer / e-commerce DTC
Ch. 71 — Arts/Entertainment/Recreation	Done (Batch 2) — 2 sub-clusters: fitness studio / attractions
Ch. 81 — Other Services	Done (Batch 2) — 3 sub-clusters: auto repair / personal care / funeral
Ch. 62 — Healthcare & Social Assistance	Done (Batch 2) — 4 sub-clusters: medical / dental / mental health / child care

What is queued for Batches 3-5

Ch. 23 — Construction	Batch 3 — Phase 2 — residential trades, commercial GCs, civil/heavy
Ch. 54 — Professional Services	Batch 3 — Phase 3 — legal, accounting, A&E;, consulting, vet, photo
Ch. 53 — Real Estate & Rental	Batch 3 — Phase 3 — agents, property mgmt, self-storage

Ch. 52 — Finance & Insurance	Batch 3 — Phase 3 — banking, insurance agencies, RIAs
Ch. 31-33 — Manufacturing	Batch 4 — Phase 4 — biggest sector, 346 codes (food/bev, industrial, advanced)
Ch. 42 — Wholesale Trade	Batch 4 — Phase 4 — distributors and rep firms
Ch. 48-49 — Transportation/Warehousing	Batch 4 — Phase 4 — trucking, transit, courier, warehouse
Ch. 56 — Admin & Support / Waste	Batch 4 — Phase 4 — staffing, janitorial, pest, security, waste
Ch. 51 — Information	Batch 5 — Phase 5 — SaaS, media, telecom, publishing
Ch. 61 — Educational Services	Batch 5 — Phase 5 — tutoring, K-12, bootcamps, trade schools
Ch. 55 + 92 — Out of scope	Batch 5 — Friendly rejection logic for holding co + public admin
Part 3 — NAICS-4 Failure-Rate Database	Batch 5 — 300+ codes with year-1/3/5/10 survival data
Part 4 — Implementation Roadmap	Batch 5 — 5 phases with explicit gates and engineering tasks
Part 5 — Appendices	Batch 5 — NAICS map, data sources, code structure, glossary, classifier keywords

How to continue the build

Reply with 'go batch 3' and the next slice — Construction + Professional Services + Real Estate + Finance/Insurance — will be appended. Batch 4 covers the B2B heavy roster (Manufacturing, Wholesale, Transportation, Admin/Waste). Batch 5 wraps with Information, Education, out-of-scope sectors, and the failure-rate database, implementation roadmap, and appendices.

PART 2 — SECTOR CHAPTERS

Per-sector specifications

Each chapter below is structured identically: business-model classification, sub-clusters, data sources, review taxonomy, opportunity playbook, specificity rules, cost scale, persona schema, cycle model, issue library, report sections, KPIs. Engineers should treat each chapter as a fillable template for the `sector_profiles.js` file. Anything marked 'not applicable' is a deliberate engineering decision — not an oversight.

Reading order recommendation

Read the chapter that maps to your current implementation phase. Phase 0 starts with Hospitality (Ch. 72) — the v3 sweet spot — as a refactor exercise. Phase 1 adds Healthcare (Ch. 62), Personal Care + Auto Repair (Ch. 81), Retail (Ch. 44-45), and Recreation (Ch. 71). Phase 2 moves to Construction trades (Ch. 23). Phase 3 tackles Professional Services (Ch. 54). Phase 4 handles B2B (Ch. 31-33, 42, 51). Phase 5 covers specialty branches (Agriculture, Mining, Utilities, Real Estate, Education).

CHAPTER 11 — AGRICULTURE, FORESTRY, FISHING, HUNTING

Commodity producers, ag-tourism operators, and forestry / fishing operations. The hospitality pipeline does not work because most of these have zero Google reviews and sell to processors, not consumers.

Covers 64 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.

The v3 problem in this sector

A 1,200-acre soybean operation has no Google reviews, no walk-in customers, and no nearby competitors in the hospitality sense. v3 produces an empty report. v4 splits this sector by output channel: commodity producers vs direct-to-consumer ag operations.

Sub-clusters in this sector

This sector splits into 3 sub-clusters because the business models inside diverge enough that one profile cannot serve them all. Each sub-cluster gets its own complete spec below.

Business Model	Business Description	Output Channel	NAICS Codes
agriculture.commodity	Commodity producers (crops, livestock, dairy, aquaculture)	B2B_regional	111110, 111150, 112111, 112120, 112511 — and the rest of 111xxx / 112xxx
agriculture.agritourism	Agritourism, U-pick, farm stands, agritainment	B2C_walk_in	111219, 111421, 111998, 112910 (apiculture for honey DTC) — usually rolled up to parent farming code
agriculture.forestry_fishing	Forestry, logging, commercial fishing, support services	B2B_regional	113110, 113210, 113310, 114111, 114112, 114119, 114210

11.commodity — Commodity producers (crops, livestock, dairy, aquaculture)

Business model: B2B_regional | **Profile ID:** agriculture.commodity

Covers NAICS: 111110, 111150, 112111, 112120, 112511 — and the rest of 111xxx / 112xxx

Modern formats (Part 2): Mushroom Farm (Specialty), Microgreens Producer, Hydroponics, Hemp Farm

1. Data sources (replaces v3 hospitality stack)

USDA QuickStats	NASS	County yields, prices received, planted acres	Free, public	USDA AgCensus 5-yr
USDA AMS market reports		Daily commodity prices, futures basis	Free	CME settlements
USDA FSA office directory		Local FSA agent + named programs (CSP, EQIP, CRP)	Free	Extension office
Land grant extension service		Named extension agent + variety trial results	Free	University ag dept
Farm Service Agency loan officer		Named local FSA loan officer	Free	Farm Credit branch
Ground-truth elevator list	grain	Named local elevators + cash bids	Scrape / API	Direct call
LandWatch AcreValue	/	Comparable land values + rental rates	Subscription	County assessor
NOAA / USDA RMA crop weather		Heat units, rainfall vs normal, drought monitor	Free	NWS station

2. Competitor radius

Three radii: (1) within 25 miles for shared elevators/processors and labor pool, (2) within 100 miles for buyer competition, (3) state-wide for commodity-class peers. v3's 2.5km is meaningless here.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Yield vs county average — quantitative, not review-based
- Soil health trajectory — soil tests over time
- Crop quality / grade — protein, falling number, test weight
- Labor reliability — H-2A retention, local crew availability
- Equipment uptime — downtime hours per season
- Buyer relationships — repeat business with named processors
- Insurance / risk profile — RMA loss ratio over 5 years
- Conservation participation — CSP / EQIP enrollment status

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **18** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Specialty / heirloom premium varieties	Named varieties with documented price premiums in this region	Switch 200 acres of #2 yellow corn to Floriani Red Flint — sells to Bob's Red Mill at \$0.42/lb premium vs \$0.18/lb commodity
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Organic / Real Organic / Regenerative cert	Cert body + transition timeline + premium captured	Real Organic Project certification — 12-month transition — \$0.85/bu premium on soybeans into Patagonia Provisions supply chain
Direct-to-processor named contract	Named processors within 200 miles seeking your commodity	Anheuser-Busch Wisconsin Rapids barley contract — locked basis, \$0.50/bu over CBOT
DTC / farm-stand pivot for a portion	Highway frontage, traffic counts, complementary local destinations	5% of acreage to pumpkins / sweet corn for \$40k stand revenue Oct-Dec on US-151 frontage
CSA / share program	County household income + organic demand + delivery zones	60-share CSA — Madison drop point — \$700/share — \$42k revenue from 2 acres
Regen ag certification + carbon credit	Indigo, Nori, Truterra, Bayer ForGround pricing	Truterra Sustainability Insetting — \$20/acre baseline plus stacking with Climate-Smart commodity
Conservation easement income	NRCS Wetland Reserve / state ag easement programs	WI Knowles-Nelson stewardship easement — \$2,400/acre on 80 marginal acres = \$192k one-time
Co-op participation upgrade	Named regional co-op + member benefits unused	CHS Inc patronage — increase share purchase to qualify for top-tier dividend rate
Variety trial partnership	Land-grant university researcher + on-farm trial program	UW-Madison soybean variety trial — \$2k stipend + free seed + first access to released line
Custom farming for a neighbor	Aging farmer within 10 miles + idle equipment hours	Custom-farm 600 acres for Smith Bros — \$80/ac corn, \$65/ac beans = \$87k margin on idle equipment
Equipment-share co-op	Named neighbors with overlapping but non-conflicting needs	4-farm machinery co-op for 16-row planter — split \$480k equipment \$120k each vs solo
Cover crop seed sales	Local seed dealer relationships + cover crop adoption rates	Add cereal rye production for sale to Albert Lea Seed — \$12/bu vs commodity \$4.50
Custom hay / forage premium	Named horse barns / dairies needing high-RFV hay	180-RFV alfalfa to named horse boarding barn — \$375/ton vs commodity \$180/ton
On-farm storage premium capture	Storage capacity vs production + basis seasonal pattern	Add 50,000 bu bin — capture \$0.40/bu post-harvest basis recovery = \$20k/yr at 6% borrowing cost
Trucking / logistics in-house	Per-mile cost vs paid hauling + return trip backhauls	Buy used semi for \$45k — eliminate \$0.18/bu hauling on 90k bu = \$16k/yr after fuel + repairs

Solar lease income	Utility solar developers + capacity zone in interconnect queue	Alliant Energy 80-acre solar lease — \$1,200/ac/yr vs \$185/ac/yr cash rent on marginal land
Hunting / recreational lease	Named outfitters + waterfowl flyway / deer harvest density	Pheasants Forever guided hunt lease — 320 acres CRP — \$3,500/season + 6-day hunting rights
Succession transition plan	Named ag attorneys + trust structures + Section 2032A	Iowa State CSBE estate plan — qualified family farm — \$4.5M estate tax savings via 2032A

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named buyer must be a real grain elevator, processor, or co-op verifiable in USDA listings or state ag department
- Every named program must be a real USDA, NRCS, or state program with a current sign-up window
- Every variety / cultivar must be a real released line with documented yield data
- Every named extension agent / FSA officer must be a real person currently in that role
- Every named buyer contract must reference current published basis or the commodity's price discovery point (CBOT, KCBT, etc.)

6. Cost scale (L5 — per-sector currency bands)

Farm operations spend across a wider range than hospitality. The cost scale is calibrated to typical farm operating budgets:

1 — Free or under \$500	Sign up for a program, change a marketing decision, switch a buyer
2 — \$500 to \$5,000	Soil tests, cover crop seed, fence repair, small equipment add-on
3 — \$5,000 to \$50,000	Variety switch on portion of acres, on-farm storage retrofit, cert transition
4 — \$50,000 to \$250,000	New grain bin, used semi, small DTC building, organic transition
5 — Over \$250,000	Major equipment, conservation easement (positive cashflow), land purchase

7. Persona schema (L6)

Schema: B2B_buyer (the customer is a procurement manager at a processor / elevator / co-op, not a consumer)

Required persona fields:

- Buyer name + role + company (e.g. "Tom Reynolds, grain merchandiser, Cargill Soy Crush, Cedar Rapids")
- Volume needed annually + delivery window
- Quality specs that drive premiums (protein, oil, FFA, mycotoxin)
- Decision criteria order (price, reliability, quality, location, relationship)
- Contract structure preferred (forward, basis, no-price-established)

- Pain points the buyer experiences with current suppliers
- How this farm differentiates vs the buyer's other suppliers

Example persona: Tom Reynolds, grain merchandiser, Cargill Soy Crush — Cedar Rapids

Buys 4M bushels of #1 soy/yr within 90mi. Pays \$0.10/bu premium for protein > 35.5% delivered Aug-Oct. Pain point: small farms don't test protein, so he assumes commodity grade. Differentiation: send protein test results with each load — qualifies for the premium tier automatically. LTV at this farm: 60,000 bu/yr × \$0.10 = \$6,000/yr in premiums alone.

8. Cycle model (L7 — replaces calendar seasons)

Farm cycles are driven by crop calendar, not tourist season. Each entry below maps a window to specific marketing / operational actions.

Pre-plant (Feb-Apr)	Seed orders + crop insurance deadline	Lock variety contracts, finalize FSA acreage, complete RMA elections
Plant + early season (Apr-Jun)	Cash flow trough	Operating loan draws, custom planter scheduling, in-season N decisions
Mid-season (Jun-Aug)	Crop scouting + side-dress + yield projection	Forward sell 30-50% of expected production at favorable basis windows
Pre-harvest (Aug-Sep)	Combine prep + storage + trucking	Lock harvest hauling rates, secure custom combine if needed, line up labor
Harvest (Sep-Nov)	Peak revenue + peak labor + peak risk	Direct-ship contracted bushels, hold uncontracted in on-farm storage for basis recovery
Post-harvest marketing (Nov-Mar)	Basis recovery + tax-year decisions	Sell stored grain on basis recovery, prepay inputs for next year, fund retirement plans
Off-season agritourism (Oct-Dec, Apr-May)	Pumpkin / Christmas tree / U-pick windows	Peak DTC revenue if a portion of operation is direct-to-consumer

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Weather risk dominates outcomes — mention but never lead the report with it
- Commodity price exposure — universal — mention via marketing plan, not as a "fix"
- Labor shortage — universal across H-2A users — frame as opportunity for automation / housing
- Succession / transition stress — universal in farms aged 55+ — frame as planning opportunity
- Equipment downtime during critical windows — universal — frame as backup-machinery / co-op
- Buyer concentration risk — universal — frame as diversification opportunity
- Land rent inflation outpacing yield gains — universal — frame as conservation / specialty pivot
- Carbon / nitrate regulatory pressure — emerging — frame as cover crop / 4R nutrient planning

10. Report sections (L9 — section list and order)

- 1 **Operation snapshot** — Acres, crops, livestock head, named buyers, 5-yr yield trend vs county
- 2 **Top 5 unique opportunities** — Drawn from the 18 categories above — at least 3 must be premium-capture, not cost-cut
- 3 **Buyer / processor map** — Named buyers within 200mi + unused contracts + premium opportunities
- 4 **Program / grant pipeline** — Named USDA + state programs with deadlines and dollar amounts
- 5 **Cycle action plan** — 7-window calendar with specific actions per window
- 6 **Risk register** — Named risks with mitigation, ranked by farm-level \$ exposure
- 7 **Succession / transition flag** — If operator > 55, surface a "talk to an ag estate attorney" line item
- 8 **Known industry challenges** — Weather / price / labor / regs — collapsed bottom drawer

Key performance indicators worth tracking

- Yield index vs 5-year county average (Bushels/ac actual ÷ county avg)
- Premium captured per bushel (\$/bu over commodity board price)
- Direct-to-consumer revenue as % of total (target: 5-15% to diversify)
- Conservation / program income as % of total (target: 5-20%)
- Operating expense ratio (operating cost ÷ gross revenue, target < 0.75)
- Working capital months (current assets ÷ monthly burn, target > 6 mo)

11.agritourism — Agritourism, U-pick, farm stands, agritainment

Business model: B2C_walk_in | **Profile ID:** agriculture.agritourism

Covers NAICS: 111219, 111421, 111998, 112910 (apiculture for honey DTC) — usually rolled up to parent farming code

Modern formats (Part 2): Christmas Tree Farm, U-Pick Farm / Pumpkin Patch, Maze / Agritourism Farm, Honey / Beekeeper

1. Data sources (replaces v3 hospitality stack)

Google Reviews	Places /	Standard hospitality review data	\$0.02	Yelp scrape
TripAdvisor		Family-trip reviews + ranking	\$0.01	Claude search
State directory	agritourism	Named circuits + member benefits	Free	Tourism board
Eventbrite / Facebook Events		Local fall / spring event calendar	Free tier	Manual scrape
School directory	district	Named local schools for field trips	Free	County education dept
Wedding venue listing sites		TheKnot, WeddingWire ranking	Free tier	Direct search
NOAA frost / pest data		Bloom + harvest window prediction	Free	Extension office
Census Bureau county demographics		Family income + young children density	Free	Claude general

2. Competitor radius

30 miles for U-pick / pumpkin patch / Christmas tree (consumers will drive). 60 miles for destination agritourism (corn maze, fall festival, weddings). 100 miles for unique experiences (lavender field, sunflower photography).

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Family / kids friendliness
- Photo quality of experience (Instagrammability)
- Animal interactions
- Parking + access
- Pricing transparency
- Cleanliness of restrooms / facilities
- Wait times for popular activities

- Quality of seasonal product (apples, pumpkins, trees)
- Educational content (school field trips)

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **17** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

School field trip program	Named local school districts + curriculum tie-ins	Sauk County School District 4th grade pollinator unit — 12 buses × 50 kids × \$8 = \$4,800 per Sept-Oct
Wedding / event venue licensing	County zoning + state ag tourism laws	Restored barn — 12 weekends × \$4,500 = \$54k at 0% incremental food cost (caterer brings food)
Birthday party packages	Local kids party benchmark prices	\$320 package — 10 kids — pony ride + maze + pumpkin — books out 20 weekends
CSA / pickup-share	County household demand + delivery zones	40-share Tuesday pickup CSA — \$650/share — \$26k from 1 acre — drives traffic to farm stand
Influencer hosting program	Named regional Instagram + TikTok creators	Comp 5 trips to @MidwestMomsAdventures (180k followers) — 1 post = ~3,000 farm visits
Photography session licensing	Named local photographers + sunflower / fall / Christmas tree photo demand	\$50/session photo permit — 60 sessions/season × \$50 = \$3k zero-effort revenue
Value-added product line	On-farm kitchen license + named retail outlets	Apple butter, jam, honey — sold on-farm + at 4 local Whole Foods stores via consignment
Goat yoga / farm yoga	Local yoga instructors + Instagram demand	\$28/person × 25 people × 18 sessions/yr = \$12,600 with farm-side pop-up bar add-on
Farm-to-table dinner series	Local chef partnerships + ticket benchmark	\$95/person × 40 people × 6 dinners/yr = \$22,800 — chef takes 50% of food + 100% of tips
Holiday season expansion (Christmas)	Tree species + cut-your-own market	Add wreath-making class + hot cocoa + tractor ride — extends Nov-Dec season + \$18 incremental/family
Off-season Airbnb / glamping	Yurts / tiny homes / cabin permitting + nightly rate benchmark	2 yurts × \$145/night × 180 nights = \$52k off-season revenue
Agritourism circuit partnership	Named state circuit + cross-promotion mechanics	WI Cheese / Christmas Tree / Pumpkin trail — listed on state tourism map = +30% traffic during season
Mobile e-commerce on-farm	Square / Shopify POS + barcode SKUs	iPad checkout + curbside pickup widget — 22% lift in average ticket vs cash-only

Loyalty / season pass	Local family pass benchmarks (zoo, museum)	\$95 family season pass — unlimited fall visits — 320 sold = \$30k upfront cash + repeat traffic
Festival sponsorship slot	Named regional brands seeking family-event sponsorship	Culver's "Family Saturday" sponsorship — \$4k for branded signage + free butter burgers
Beekeeping educational program	State 4-H + extension partnerships	\$45/person × 12 people × 4 workshops/yr = \$2,160 + sells \$90/jar honey to attendees
Wellness retreat / wedding shower hosting	Named wedding planners + retreat centers	Bridal shower package — \$850 for 20 — 18 weekends/yr = \$15,300

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named partner (school, photographer, brand) must be verifiable in Google or state directory
- Every event recommendation must include attendance benchmark from comparable farm in same state
- Every program (CSA, season pass) must include a real benchmark price from an existing operation
- Photography licensing must reference real per-session benchmark in this region

6. Cost scale (L5 — per-sector currency bands)

Agritourism cost scale is closer to small retail / hospitality:

1 — Free or under \$200	Sign up for circuit, post on Facebook, comp influencer trip
2 — \$200 to \$2,000	Signage, photo permits, simple event hosting, basic marketing
3 — \$2,000 to \$20,000	Yurt / tiny home, wedding-prep barn cleanup, POS system
4 — \$20,000 to \$100,000	Major facility upgrade, commercial kitchen, parking expansion
5 — Over \$100,000	Wedding barn restoration, lodging buildout, large equipment

7. Persona schema (L6)

Schema: B2C_consumer (use the existing v3 hospitality persona schema)

Required persona fields:

- Name + age + family composition
- Distance willing to drive for this experience
- Spend trigger (what makes them open wallet)
- 5-star trigger (what earns the great review)
- Photo behavior (do they share on Instagram / TikTok)
- Word-of-mouth trigger (what makes them tell other parents)
- Lifetime value (visits per year × ticket × food / retail attach)

- Acquisition channel (which named Facebook group / school)

Example persona: Megan, 36, Madison mother of 2 kids ages 4 and 7

Drives 45 min for a "good" fall experience. Spend trigger: pumpkin painting station + free hot cider + Instagrammable hay bale photo wall. 5-star trigger: clean restrooms + short maze line + tractor ride that feels safe. Tells 4-6 other parents in her school WhatsApp group. LTV: 2 visits/yr × \$48 (admission + pumpkin + concessions) = \$96/yr × 6 yrs while kids are young = \$576.

8. Cycle model (L7 — replaces calendar seasons)

Agritourism runs on commodity-product seasonal calendar — different from generic tourism:

Pre-bloom (Mar-Apr)	Spring photo bookings + maple syrup	Open early for spring photos, market maple syrup festival
Bloom + spring (May-Jun)	School field trip peak	Drive school district sign-ups, run flower-photo days
Pick-your-own spring (Jun-Jul)	Strawberry / cherry season	Heavy social media on bloom + harvest dates, partner with bakers
Summer dead zone (Jul-Aug)	Heat + competing summer activities	Goat yoga, sunflower fields, farm-to-table dinners — lifestyle pivot
Fall peak (Sep-Oct)	Apples, pumpkins, mazes — 60% of annual revenue	Maximize daily ops, weekend volume, school weekday traffic
Holiday (Nov-Dec)	Christmas trees + wreaths + cocoa	Cut-your-own + ornament shop + photos with Santa
Off-season (Jan-Feb)	Wedding bookings + planning + Airbnb	Wedding venue tours, glamping, planning next year

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Weather cancellation — universal — refund policy + rain check + insurance
- Parking overflow on peak weekends — universal — early-arrival incentive + shuttle
- Restroom complaints — universal — invest before season
- Animal smell / safety concerns — universal — signage + handwashing stations
- Pricing surprise (gates + activities + photos add up) — frame as bundled pass to remove
- Long lines at popular activities — frame as timed-entry / virtual queue opportunity

10. Report sections (L9 — section list and order)

- Top 5 unique experiences to add** — Drawn from 17-category playbook
- Quick-win operational improvements** — Under \$2k actions
- Wedding / events licensing roadmap** — If applicable based on county zoning
- Agritourism circuit + partnership map** — Named partners within 60 miles

- 5 **Customer personas (4 personas + lost customer)** — Family / school group / wedding party / commercial buyer
- 6 **Seasonal cycle plan** — 7-window calendar specific to this farm's product
- 7 **Known operational challenges** — Weather / parking / restrooms — collapsed drawer

Key performance indicators worth tracking

- Avg ticket per family (admission + add-ons + food + retail)
- Photo-share rate (Instagram tags per 100 visitors)
- Repeat visit rate within season
- Wedding revenue as % of total (target: 15-30% for ag-tourism + wedding)
- School field trip bookings per fall season
- Off-season revenue % of annual (target: > 10% to smooth cash flow)

11.forestry_fishing — Forestry, logging, commercial fishing, support services

Business model: B2B_regional | **Profile ID:** agriculture.forestry_fishing

Covers NAICS: 113110, 113210, 113310, 114111, 114112, 114119, 114210

Modern formats (Part 2): Aquaculture / Fish Farm, Oyster / Shellfish Farm

1. Data sources (replaces v3 hospitality stack)

State DNR forestry program	Named foresters + cost-share programs	Free	Extension office
USDA Forest Service	National forest sale schedule + bid history	Free	BLM equivalent
NOAA Fisheries / state fish & wildlife	Quota allocation + stock status + permits	Free	Industry assoc.
Random Lengths / Madison's Lumber	Pricing for lumber, pulp, chips	Subscription	TSX-listed firms
ATFS / FSC / SFI cert bodies	Cert status + premium markets	Free public	Direct cert auditor
LandWatch + state timber sale lists	Comparable timberland values + sale prices	Free / Sub	County assessor
USDA APHIS	Named pests + quarantine zones (e.g. spongy moth)	Free	State ag dept

2. Competitor radius

Forestry: 100 miles to mill / paper plant. Logging: 60 miles for daily haul. Commercial fishing: defined by quota zone + fishery management area, not geographic radius.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Stumpage price captured vs regional benchmark
- Cert status (FSC, SFI, ATFS) and price premium realized
- Mill / processor relationships
- Equipment uptime + crew reliability
- Safety record (logging is highest-fatality industry)
- Habitat / regulatory compliance record

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **12** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Named cert pursuit (FSC / SFI / ATFS)	Cost of certification + premium captured in this region	FSC certification — \$3,500 audit + \$0.50/MBF year-1 — captures premium from named furniture maker
Carbon credit registration	Climate Action Reserve / ACR / VCS protocols	CAR Improved Forest Management — \$8/ton × 0.5 ton/ac = \$4/ac/yr passive income on managed acres
Conservation easement income	State + USDA forest legacy programs	Forest Legacy easement — \$1,200/ac one-time payment retains ownership + working forest
High-value species premium	Named buyers seeking veneer-grade hardwood	Roy Anderson Lumber Wisconsin — pays \$1,800/MBF for #1 sugar maple veneer vs \$480 for sawlog
Logging contract diversification	Mill, pulp plant, biomass facility within 100mi	4-buyer contract mix (mill + pulp + biomass + firewood) reduces single-buyer concentration risk
Recreational lease income	Hunting club + outfitter demand in region	600-acre hunting lease — \$14,000/yr to named local outfitter — zero impact on timber rotation
Conservation cost-share enrollment	NRCS EQIP forestry practice list + state programs	EQIP timber stand improvement — \$185/ac on 80 ac = \$14,800 cost-share for marked thinning
Timber stand inventory + management plan	State-licensed forester + cost-share for plan	WI DNR cost-shared forest management plan — \$0 to landowner — unlocks \$14k of EQIP funds above
Niche product (firewood, pellets, mulch)	Local retail demand + bagging / delivery economics	Bagged firewood to 4 named gas stations — \$7/bag × 800 bags/mo = \$5,600/mo from cull material
Custom milling for craftsman market	Wood-Mizer dealer + local custom-furniture market	Custom-mill 80 hardwood logs/yr for named cabinetmakers — \$1.20/BF retained vs \$0.48 commodity
Education / workshop hosting	Land trust + university extension partnerships	Aldo Leopold Foundation forester-day on this property — \$1,500 honorarium + reputation lift
Storm-damage / salvage rapid response	Named ins. adjusters + emergency services list	Pre-positioned on insurance approved-vendor list — captures storm-salvage premium \$/MBF

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named buyer must be a real mill or processor in state forester directory
- Every cost-share program must reference current sign-up window with state forester contact
- Every cert body must be a real ANSI-accredited certifier
- Every species-grade premium must reference current published pricing

6. Cost scale (L5 — per-sector currency bands)

Forestry operates on multi-decade cycles. Cost scale is calibrated to a 5-year planning horizon:

1 — Free or under \$1,000	Sign up for program, get cost-shared management plan
2 — \$1,000 to \$10,000	Cert audit, inventory work, small equipment
3 — \$10,000 to \$100,000	Forwarder / skidder used, sawmill, custom milling shop
4 — \$100,000 to \$1M	New harvesting equipment, pellet mill startup
5 — Over \$1M	Mill expansion, large land acquisition

7. Persona schema (L6)

Schema: B2B_buyer (mill procurement, pulp procurement, custom-furniture buyer)

Required persona fields:

- Mill or processor name + procurement officer
- Species + grade demand + annual volume
- Cert preference (FSC / SFI / ATFS)
- Payment terms + scaling method
- Geographic supply shed they pull from
- Long-term agreement appetite

8. Cycle model (L7 — replaces calendar seasons)

Forest cycle is dual: short cycle (annual market) + long cycle (rotation length). Both surface in the report:

Winter (Dec-Mar)	haul	Frozen ground + access to wet sites	Schedule 60% of year's harvest in frozen-ground months
Spring (Apr-May)	breakup	Mud + access closures	Crew off-time for equipment maintenance + planning
Summer (Jun-Aug)		Dry ground + fire season	Ridge / sandy site harvests, fire-management precautions
Fall (Sep-Nov)		Pre-freeze prep + planting + cert audits	Plant seedlings, schedule cert recertification
5-yr inventory cycle		Re-cruise and update plan	Full timber inventory + management plan refresh
Rotation (40-100 yrs)	cycle	Final harvest + regen	Site preparation, replanting, post-harvest residual sale

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Safety incidents — fatality rate is high — universal
- Weather-driven access loss — universal — frame as winter-haul preparation
- Stumpage price volatility — universal — frame as buyer diversification
- Pest / disease outbreak (spongy moth, EAB, oak wilt) — frame as proactive cert + management
- Labor shortage in skilled forestry crews — universal — frame as crew retention / training
- Insurance availability for logging contractors — frame as safety record / cert as differentiator

10. Report sections (L9 — section list and order)

- 1 **Tract / operation snapshot** — Acres, species mix, current rotation stage, cert status
- 2 **Top 5 unique opportunities** — Drawn from 12-category playbook
- 3 **Buyer / mill map** — Named buyers within 100 miles + species fit
- 4 **Cert + carbon roadmap** — Path to FSC/SFI/ATFS + carbon registration
- 5 **Cost-share + program inventory** — EQIP, state forestry, Forest Legacy with dollar amounts
- 6 **Cycle plan** — Annual + rotation calendar with action items
- 7 **Risk register** — Pest, fire, market, regulatory
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Stumpage \$/MBF captured vs regional benchmark
- Cert premium realized (\$/MBF over commodity)
- Carbon income per acre per year
- Recreational lease income per acre per year
- Days-in-season equipment utilization (target > 180/yr)
- Safety incidents per million hours worked (target = 0)

Out-of-scope NAICS codes inside this sector

These codes belong to this sector but BizRadar refuses to analyze them — either because they are not real businesses, are regulated monopolies BizRadar cannot help, or do not have customer-facing operations to optimize.

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Support activities for ag/forestry — analyzed under their parent (e.g. crop harvesting service rolls into agriculture.commodity profile)

Engineering note for this sector

Sector 11 is the first sector where the v3 hospitality pipeline produces obviously broken output. Engineers should expect to spend ~3 weeks on this sector alone — most of the work is sourcing real producer / buyer / program directories rather than writing code. Build the agriculture.agritourism profile first (closest to v3) then commodity, then forestry.

CHAPTER 21 — MINING, QUARRYING, OIL & GAS EXTRACTION

Industrial extraction operations selling to refiners, smelters, construction. Heavy regulation, capital-intensive, multi-year decision cycles.

Covers 21 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.

The v3 problem in this sector

A limestone quarry has zero Google reviews and competes within a 200-mile freight radius for asphalt and concrete plants. v3's "competitors within 2.5km" returns nothing useful and the report is empty.

Sub-clusters in this sector

This sector splits into 2 sub-clusters because the business models inside diverge enough that one profile cannot serve them all. Each sub-cluster gets its own complete spec below.

mining.aggregates	Aggregates, sand, gravel, dimension stone, industrial minerals	B2B_regional	212311-212390 (stone, sand, gravel, clay, kaolin)
mining.oil_gas_metal_s	Oil and gas extraction, coal, metal mining, support services	B2B_national	211120, 211130, 212114-212290, 213111-213115

21.aggregates — Aggregates, sand, gravel, dimension stone, industrial minerals

Business model: B2B_regional | **Profile ID:** mining.aggregates

Covers NAICS: 212311-212390 (stone, sand, gravel, clay, kaolin)

1. Data sources (replaces v3 hospitality stack)

USGS Mineral Industry Surveys	Production volumes + prices by commodity	Free	State geological survey	
MSHA Data Retrieval System	Safety record by mine ID + competitor records	Free	OSHA equivalent	
EPA ECHO	Environmental compliance + violations history	Free	State EPA	
State permit databases	Reclamation bond + permit status	Free	Direct dept call	

NRMCA / NSSGA member directory	Named ready-mix + asphalt buyers	Sub / member	Industry assoc.
DOT contract awards	Public-works aggregate contracts	Free	State DOT
BlueBook / D&B;	Buyer credit + payment behavior	Subscription	Industry references
LandWatch + USGS quad maps	Reserve life + adjacent reserves	Free / Sub	Geologist consult

2. Competitor radius

Aggregates do not freight far. Default radius: 50 miles for ready-mix / asphalt customers; 100 miles for high-spec aggregates (rip-rap, ballast); 200+ miles for high-value industrial minerals (silica, kaolin).

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- On-time delivery rate to ready-mix plants
- Material spec compliance (ASTM C33, AASHTO M80)
- Truck cycle time at scale
- Account payment terms vs competitors
- Safety record (MSHA reportable incidents)
- Reclamation / community standing

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **12** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

DOT prequalification + bid pursuit	State DOT certified producer list + naming an account manager	Wisconsin DOT certified aggregate producer — \$0 application — qualifies for \$80M annual road program bidding
Specialty product line addition	Premium spec products (rip-rap, decorative, road base subgrade)	Add limestone rip-rap production — \$14/ton vs \$7/ton commodity gravel — captures DNR shoreline projects
Long-term offtake with ready-mix	Named ready-mix accounts within 50 miles	5-yr supply agreement with Cemex Madison plant — \$2.4M/yr at indexed pricing
Reserve expansion / next-pit permitting	Adjacent landowners + state mining permit timeline	Lease 80-acre adjacent reserve from named landowner — \$1,200/ac signing + \$0.45/ton royalty = 25-yr reserve life extension
Reclamation as community-asset narrative	County land trust + post-mining-use plans	Post-mining quarry-as-recreational-lake plan — community PR + state cost-share for closure activities
Named ESG reporting + certifications	ISO 14001 + WBA Stone Standard + EPD program	NSSGA Stone Standard Tier-2 certification — qualifies bids on green-building specs (USGBC LEED v4 sourcing credit)

Bagged / DTC consumer channel	Big-box retailer (Home Depot, Menards) supply specs	Add bagged decorative stone — Menards regional supply contract — \$4.50/40lb bag retail
Recycled aggregate from C&D;	Demolition contractors + state recycled-content specs	Process C&D; concrete — sell at \$11/ton vs raw quarried — captures DOT recycled-content credits
Trucking-fleet integration	Owner-operator pool + mixed-fleet economics	4-truck owned fleet — captures \$0.18/ton-mile delivery margin on 200k tons/yr = \$108k
Crushing-on-demand at customer site	Mobile crusher rental + DOT highway-job market	Mobile crushing service to DOT contractors — \$2.40/ton vs hauling raw — wins by avoiding haul cost
Industrial silica / frac-sand pivot	Sand specs + named oilfield service buyers	High-purity silica for glass — named to Owens-Illinois supply list — \$42/ton vs \$9/ton construction sand
Energy mine-electrification narrative	Battery-electric haul trucks + state utility programs	Pilot 2 battery-electric haul trucks — XCel Energy rebate + ESG narrative for institutional buyers

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named buyer must be in NRMCA / NAPA / NSSGA member directory or a verified ready-mix / asphalt plant
- Every spec product must reference a real ASTM / AASHTO / state DOT specification number
- Every permit / reclamation reference must name a real state agency contact + permit number range
- Every cert pursuit must name a real auditor + member program

6. Cost scale (L5 — per-sector currency bands)

Aggregate operations are heavy-capital. Cost scale reflects industry capex norms:

1 — Under \$5,000	Cert application, DOT prequal, marketing collateral
2 — \$5,000 to \$50,000	Lab equipment, scale upgrade, ESG report, marketing video
3 — \$50,000 to \$500,000	Mobile crusher, screen plant upgrade, ready-mix lab
4 — \$500,000 to \$5M	New crusher, conveyor upgrade, reserve land lease
5 — Over \$5M	Pit expansion, electrification fleet, new product line buildout

7. Persona schema (L6)

Schema: B2B_buyer (procurement officer at named ready-mix / asphalt / concrete-block plant)

Required persona fields:

- Buyer name + role + plant location
- Annual tonnage demand + spec mix
- Decision criteria order (price, on-time, spec compliance, ESG, reliability)
- Pain points with current suppliers
- Contract structure preferred (spot, indexed, fixed)
- Audit / cert requirements

8. Cycle model (L7 — replaces calendar seasons)

Aggregate sales mirror construction season + DOT bidding calendar:

Winter (Dec-Feb)	Construction off-season + permit / DOT bid prep	Stockpile inventory, file permits, pursue DOT prequal, negotiate offtake
Spring (Mar-May)	Construction ramp-up + project starts	Activate spring delivery contracts, surge crew + trucking
Summer (Jun-Aug)	Peak construction + DOT season	Maximum production + delivery, safety + heat protocols
Fall (Sep-Nov)	Construction wind-down + winter prep	Final-month surge, fall inventory build, equipment maintenance
Rolling: DOT bid lettings	State-published bid letting calendar	Submit aggregate bids 6 weeks before letting
Annual: permit / reclamation reporting	State filing deadlines	Reclamation progress + bond review

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Community / NIMBY pushback — universal — frame as proactive community relations
- Permit + reclamation cost — universal — frame as planning + bond reduction strategies
- Truck driver shortage — universal — frame as driver retention + autonomous haul pilot
- Diesel cost volatility — universal — frame as fuel surcharge + electrification
- Reserve depletion — universal but timeline-specific — frame as adjacent leases
- Mine safety record visibility — universal — frame as MSHA-zero campaign for procurement

10. Report sections (L9 — section list and order)

- 1 **Operation snapshot** — Reserves remaining, products, named buyers, tonnage trend
- 2 **Top 5 differentiation moves** — Drawn from 12-category playbook
- 3 **Buyer + DOT pipeline** — Named accounts + DOT bid letting calendar
- 4 **ESG / cert roadmap** — Named cert + premium realization plan
- 5 **Permit + reclamation calendar** — Filing deadlines + bond optimization
- 6 **Fleet / equipment plan** — Replacement timing + electrification pilots
- 7 **Safety scorecard** — MSHA + lost-time-incident benchmark
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Tons sold per year + average \$/ton realized
- On-time delivery rate to top 5 customers (target > 97%)
- MSHA reportable incidents per 200k hours (target = 0)
- DOT prequalification status + active bids count
- Reserve life remaining (years at current production)
- ESG cert coverage % of production (target > 50%)

21.oil_gas_metals — Oil and gas extraction, coal, metal mining, support services

Business model: B2B_national | **Profile ID:** mining.oil_gas_metals

Covers NAICS: 211120, 211130, 212114-212290, 213111-213115

1. Data sources (replaces v3 hospitality stack)

EIA (Energy Information Admin)	Production + price data by basin / county	Free	State oil + gas comm	
USGS Mineral Industry Surveys	Metal production + reserves	Free	State geological survey	
MSHA Data Retrieval	Safety + quarterly reports	Free	OSHA	
EPA ECHO + state DEQ	Environmental compliance	Free	State EPA	
SEC EDGAR (for public co)	Reserves + production reports	Free	10-K of named operator	
London Metal Exchange / NYMEX / CME	Commodity pricing + futures	Sub / public	Vendor screen	
BLM / state agency lease databases	Lease status + bonus + royalty	Free	BLM regional office	
Drilling Info / Enverus	Wells, leases, production by operator	Subscription	State commission	

2. Competitor radius

Oil + gas: basin-defined (Permian, Marcellus, Bakken, etc.). Metals: national or global by commodity. Coal: regional (Powder River, Illinois, Appalachian).

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Operating cost per BOE / ton produced
- Reserves replacement ratio
- Safety + environmental compliance record
- Joint-venture / partner reputation
- Regulator + community standing
- ESG rating from named agencies (Sustainalytics, MSCI)

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **12** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

ESG / sustainability disclosure upgrade	TCFD + SASB + GRI standards adoption	Adopt SASB Mining standard disclosures — captures institutional investor flow + green bond eligibility
Methane abatement narrative (oil + gas)	EPA OOOOb + state methane rules + EU CBAM exposure	OOOOOb-compliant tanks + LDAR — captures premium \$/Mcf from named LNG buyer requiring low-methane gas
Long-term offtake / royalty trust	Named refinery + smelter buyers + capital-markets structures	10-yr take-or-pay with named refinery — locks margin + qualifies for project finance
Carbon-credit registration (CCS, CCUS)	45Q tax credit + named CCS partners	45Q \$85/ton sequestered — co-develop with named midstream operator — \$24M/yr at 280k tons
Tribal / community agreements (FPIC)	Named tribal nations + ICMM principles	Free Prior Informed Consent agreement with named tribal council — unlocks \$40M federal grant for project
Battery-minerals positioning (Li, Co, Ni, Cu, REE)	IRA Section 30D battery sourcing + named OEM offtake	IRA-qualifying lithium spodumene supply to GM Ultium battery JV — \$2,600/ton vs \$1,400 spot
Critical-minerals federal program enrollment	DPA Title III + DOE LPO + DoD Industrial Base Fund	DPA Title III \$30M cost-share for high-grade graphite refining
Reclamation bond optimization	Self-bonding + corporate guarantee alternatives	Replace cash bond with corporate surety — frees \$14M working capital at 1.5% premium cost
Mineral lease consolidation	Adjacent operator landman + landowners + JV structures	Consolidate fragmented leases via 2-mile lateral pooling — increases recoverable reserves 18%
Royalty-and-streaming financing	Wheaton / Franco / Royal Gold / private royalty buyers	\$60M streaming deal with named royalty company — non-dilutive growth capital at 5-7% cost
Tour + community education program	County school district + state museum partnerships	Quarterly mine tour + scholarship program — addresses local opposition + builds talent pipeline
Tailings repurposing / secondary recovery	Tailings reprocessing tech + secondary metal markets	Reprocess 800-acre tailings pile for residual copper — \$4M capex + \$1.8M/yr revenue + reclamation credit

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named buyer must be a real refinery / smelter / OEM / utility
- Every cert / standard must reference real body (SASB, ISO 14001, IRMA)
- Every program must reference real federal agency and program name
- Every cost / royalty figure must reference public benchmark (EIA, USGS, S&P;)
- Every tribal / community agreement must reference real nation and contact

6. Cost scale (L5 — per-sector currency bands)

Oil/gas/metals operate at industrial capex scale. Bands are calibrated accordingly:

1 — Under \$50,000	Cert application, ESG report, community PR
2 — \$50,000 to \$500,000	Methane LDAR, stakeholder engagement, study
3 — \$500,000 to \$5M	Equipment retrofit, expanded compliance, pilot project
4 — \$5M to \$50M	Major equipment, capacity expansion, JV stake
5 — Over \$50M	New mine, refinery, processing facility

7. Persona schema (L6)

Schema: B2B_buyer (refinery, smelter, OEM procurement, utility)

Required persona fields:

- Buyer org + named procurement / commercial contact
- Volume needed + spec window
- Cert / ESG requirements
- Term length preference + renewal options
- Pain points with current supply portfolio
- Strategic priorities (security of supply, ESG, cost)

8. Cycle model (L7 — replaces calendar seasons)

Industrial extraction runs on commodity-price + permit + capital-allocation cycles, not seasons:

Annual: AFE + capital plan	Board / partner capital allocation	Submit AFEs by Q4 for next year capital
Annual: regulator + bond review	State + federal annual filings	File reclamation + ops reports
Quarterly: SEC / SEDAR (public)	Reserve report + production update	Coordinate IR + analysts
Multi-year: permit + EIS cycle	3-7 year EIS / permit timeline	Stakeholder engagement + biological surveys
Commodity cycle (3-10 yr)	Macro price cycle peaks + troughs	Counter-cyclical capex, hedging program
Election cycle	Federal + state regulatory shifts	Trade group engagement, scenario planning

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Commodity price exposure — universal — frame as hedging + diversification

- Community + tribal opposition — universal — frame as proactive consultation
- Permit + EIS delay — universal — frame as parallel-pathing + contingency
- Reclamation cost overrun — universal — frame as bond optimization
- Worker safety + fatality risk — universal — frame as zero-harm program
- Capital availability swings — universal — frame as alternative financing
- Methane / carbon scrutiny — emerging — frame as abatement narrative

10. Report sections (L9 — section list and order)

- 1 **Asset snapshot** — Reserves, production, named buyers, ESG profile
- 2 **Top 5 strategic moves** — Drawn from 12-category playbook
- 3 **Buyer + offtake map** — Named buyers + contract status + premium opportunities
- 4 **ESG + cert roadmap** — Path to recognized ESG ratings + premium markets
- 5 **Permit + community engagement plan** — Stakeholder map + filing calendar
- 6 **Capital structure + financing options** — Streaming / royalty / project finance / federal cost-share
- 7 **Risk register** — Price, permit, community, safety, reclamation
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Operating cost per BOE / ton produced vs basin / commodity benchmark
- Reserves replacement ratio (added ÷ produced, target > 1.0)
- Total recordable incident rate (target = 0)
- ESG rating from named agency (target: top quartile)
- Methane intensity (kg CO₂e/Mcf for gas, target < 0.20)
- Community-grievance log open items (target = 0)

Engineering note for this sector

Sector 21 is rarely a v3 customer because mining + oil/gas operations have full-time strategy teams. The v4 value proposition is for small / mid-cap operators (single-mine, single-basin) who lack that strategy depth. Phase 5 work — build last unless a specific request arrives.

CHAPTER 22 — UTILITIES (ELECTRIC, GAS, WATER, SEWAGE)

Regulated monopolies and utility cooperatives. The "competition" framing is regulatory + customer-satisfaction, not market-share.

Covers 14 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.

The v3 problem in this sector

A regulated electric utility has a 100% market share in its franchise area. v3's "competitor gap analysis" is meaningless. v4 reframes the analysis as regulator + customer + grid-modernization optimization.

22.electric — Electric utilities (IOU, coop, municipal) and gas / water utilities

Business model: Regulated_monopoly | **Profile ID:** utilities.electric

Covers NAICS: 221111-221122, 221210, 221310-221330

1. Data sources (replaces v3 hospitality stack)

FERC eLibrary + Form 1	Rates, transmission, financial	Free	State PUC	
EIA Form 861 / 923	Sales, generation, customer counts	Free	Annual report	
State PUC dockets	Rate cases + complaints + IRP filings	Free	Direct PUC office	
J.D. Power utility studies	Customer satisfaction benchmarks	Subscription	ACSI alternative	
SAIDI / SAIFI public data	Reliability metrics + outage history	Free EIA / state	Internal SCADA	
NERC reliability standards	Compliance + incident history	Free public	Audit report	
NRECA / APPA / EEI	Industry benchmarks by class	Member-only	Public summary	
DOE LBNL utility cost data	O&M; and capital benchmarks	Free public	EIA	

2. Competitor radius

Same NERC region for reliability benchmarking. Same state for rate-case comparison. Same utility class (IOU vs coop vs muni) for governance benchmarking.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- SAIDI / SAIFI vs region
- Rate level vs region
- Customer service responsiveness
- Billing accuracy + clarity
- Outage communication during emergencies
- Program participation rates (DR, EE, low-income)
- Capital investment efficiency

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Time-of-use critical-peak adoption + rate	Named regulatory model + customer adoption rates	TOU rate offering — modeled on PG&E; TOU-D-PRIME — 28% bill savings for EV-owning customers
EV charging network build-out	NEVI funds + utility-owned charging program	NEVI \$5M corridor charging award + \$0.18/kWh DCFC tariff revenue
Distributed generation / community solar	Named developers + state community-solar program	Community solar program — 20MW in three sites — adds \$2.4M annual margin + low-income subscription
Demand response + virtual power plant	Named DR aggregators (CPower, Voltus, OhmConnect)	40MW residential DR via OhmConnect — \$80/kW-yr capacity payment + grid services revenue
AMI / smart meter deployment	AMI vendor mix + business case	AMI 2.0 deployment — Itron Riva — \$80M capex + \$14M annual O&M; savings + customer engagement features
Grid resilience / storm hardening	FEMA + state hardening grants + IIJA funds	IIJA Grid Resilience Innovation Partnerships \$20M award — undergrounding 60mi feeder + microgrid
Battery storage as transmission asset	FERC Order 841 + ISO market participation	40MW/160MWh battery as T&D; deferral — 4-yr defers \$30M substation upgrade + \$4M/yr energy arbitrage
Microgrid offerings to commercial	Named anchor customer (hospital, military, university)	Hospital microgrid pilot — \$14M capex + \$2.4M annual reliability premium revenue
Energy efficiency program expansion	State EE goal + named program portfolio	Expand HVAC tune-up rebate — 12k participants/yr — meets state EE goal + earns performance incentive
Low-income customer programs	LIHEAP partnership + state weatherization	CARE program partnership with state LIHEAP — 22k enrolled — bills 25% lower + reduces uncollectibles 18%

Dark-fiber leasing	Existing fiber inventory + telecom market	Lease excess dark fiber to Verizon + 3 ISPs — \$1.8M/yr non-rate revenue
Heat pump electrification campaign	Named HVAC contractor network + state rebate	Heat pump promo — partner with 14 named HVAC contractors — \$1,800 utility rebate + \$2k state — 4,000 installs/yr
Wildfire mitigation + PSPS protocol (West)	CalFIRE + state utility commission rules	Multi-year wildfire mitigation plan + PSPS protocol — recovers \$400M/yr in rate base
Rate-case PR + stakeholder engagement	Named consumer advocates + intervenor groups	Pre-filing stakeholder workshops with named advocate orgs — improves rate-case approved % by ~12 points

5. Specificity rules (L4 — what 'real named entity' means here)

- Every program must reference real state PUC docket + named regulator division
- Every grant / fund must reference real federal program (NEVI, IIJA, DOE LPO, IRA 48E)
- Every named partner / aggregator / vendor must be a real company
- Every benchmark must reference EIA, J.D. Power, EEI, or NRECA published data
- Every named ratepayer / advocate must be real intervenor in this PUC docket history

6. Cost scale (L5 — per-sector currency bands)

Utilities operate at infrastructure scale. Cost bands reflect typical capex programs:

1 — Under \$100,000	Customer comm program, study, regulator filing prep
2 — \$100,000 to \$1M	Pilot DR, small EE program, dark-fiber buildout
3 — \$1M to \$10M	Microgrid pilot, AMI software, grid-edge platform
4 — \$10M to \$100M	AMI deployment, substation rebuild, microgrid full
5 — Over \$100M	Generation asset, major T&D; rebuild, AMI 2.0 system-wide

7. Persona schema (L6)

Schema: B2B_buyer (regulator + large customer) plus B2C_consumer (residential ratepayer)

Required persona fields:

- Stakeholder type (PUC commissioner, large industrial, residential, advocate, intervenor)
- Decision authority + influence
- Top concern in their constituency / portfolio
- Past dockets / positions taken
- Engagement preference (formal filing, workshop, op-ed, direct meeting)
- Risk to utility if mishandled

8. Cycle model (L7 — replaces calendar seasons)

Utilities run on regulatory + capital + storm cycles, not calendar:

Rate-case cycle (3-5 yrs)	PUC-defined window	Pre-file workshops, file rate case, intervene, rebuttal, brief, order
Annual: IRP filing	State integrated resource plan	Long-term resource portfolio + clean-energy transition
Annual: capital plan	Board capital approval	AFE list + ROE deployment + grant pursuits
Storm season (varies by region)	Hurricane / tornado / wildfire	Pre-position crews, emergency comm protocol, mutual aid
Peak demand season	Summer (Sun Belt) / winter (Northeast)	DR activation, mutual aid agreements, operating reserves
Election cycle	State / federal regulatory shifts	Industry advocacy + commissioner education

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Reliability outages during storms — universal — frame as resilience + restoration
- Bill spikes during weather extremes — universal — frame as TOU + EE + bill protection
- Rate increases / affordability — universal — frame as program help + transparency
- Customer service hold times — universal — frame as digital self-service + AMI
- Aging infrastructure — universal — frame as grid-modernization narrative
- Wildfire liability (Western states) — universal — frame as PSPS + mitigation
- Clean-energy transition pace — varies — frame as pragmatic IRP + workforce

10. Report sections (L9 — section list and order)

- 1 **Utility snapshot** — Customer count, peak load, generation mix, recent rate-case outcome
- 2 **Top 5 strategic moves** — Drawn from 14-category playbook
- 3 **Stakeholder map** — PUC commissioners, advocates, large customers, intervenors
- 4 **Federal funding pipeline** — IIJA / IRA / DOE programs with submission calendar
- 5 **Reliability + storm response plan** — SAIDI / SAIFI targets + restoration playbook
- 6 **Customer program portfolio** — EE / DR / EV / community solar + benchmarks
- 7 **Rate-case roadmap** — Pre-filing, intervenor management, key witness list
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- SAIDI minutes (target: top quartile in NERC region)
- SAIFI events (target: top quartile in NERC region)
- J.D. Power customer satisfaction index (target: top tertile)
- EE program savings (% of sales, target meets state goal)
- Federal funding awarded (NEVI / IIJA / IRA)
- Rate-case ROE approved vs requested (% of ask)

Out-of-scope NAICS codes inside this sector

These codes belong to this sector but BizRadar refuses to analyze them — either because they are not real businesses, are regulated monopolies BizRadar cannot help, or do not have customer-facing operations to optimize.

221121	Bulk power transmission — typically operated by ISO/RTO, not a customer-facing business BizRadar can help

Engineering note for this sector

Sector 22 is the smallest customer pool (only ~3,000 utilities in the US) but they are extremely high-LTV. The v4 value prop is for small/mid coops + munis (EnergyTrust-tier customers). Phase 5 work — pursue only after a co-op trade association expresses interest.

CHAPTER 44-45 — RETAIL TRADE

Brick-and-mortar specialty retail, auto dealers, e-commerce DTC. The closest cousin to v3 hospitality, but with sub-cluster differences that matter.

Covers 57 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.

The v3 problem in this sector

A boutique gift shop fits v3 well. A car dealer needs F&I; and service-bay analysis v3 has no concept of. A pure-play DTC e-commerce brand has no physical radius and v3's "competitors within 2.5km" returns garbage. Three distinct profiles required.

Sub-clusters in this sector

This sector splits into 3 sub-clusters because the business models inside diverge enough that one profile cannot serve them all. Each sub-cluster gets its own complete spec below.

retail.specialty_brick_mortar	Specialty brick-and-mortar (clothing, gifts, books, home, sporting goods, hardware)	B2C_walk_in	448120, 449110, 449122, 451120, 459110, 459140, 459510
retail.auto_dealer	Auto dealers (new + used cars, RVs, motorcycles, boats)	B2C_appointment	441110, 441120, 441210, 441222, 441227
retail.ecommerce_dtc	E-commerce / DTC consumer brands (online-only or omnichannel)	B2C_online	454110 (electronic shopping), 459xxx (online-only specialty)

44-45.specialty_brick_mortar — Specialty brick-and-mortar (clothing, gifts, books, home, sporting goods, hardware)

Business model: B2C_walk_in | **Profile ID:** retail.specialty_brick_mortar

Covers NAICS: 448120, 449110, 449122, 451120, 459110, 459140, 459510

Modern formats (Part 2): Independent Bookstore, Gift Shop, Toy Store, Specialty Coffee Roaster Retail, Boutique Apparel

1. Data sources (replaces v3 hospitality stack)

Google Places + Reviews	Reviews, photos, hours	\$0.02	Yelp
Yelp Fusion API	Reviews + categories	\$0.01	Manual
Foursquare SafeGraph /	Foot traffic + dwell time	Subscription	Door counter
Local chamber of commerce	Named local events + co-op opportunities	Free	Direct call
Shopify / Square retail benchmarks	Industry conversion + ATV data	Subscription public /	NRF reports
Etsy + Faire wholesale	Maker / vendor sourcing	Free / sub	Trade show
Mailchimp / Klaviyo retail benchmarks	Email open / conversion benchmarks	Free public	Internal CRM
Local mall / district leasing data	Foot traffic + co-tenant mix	CoStar / direct	Property mgr

2. Competitor radius

0.5 km urban / mall, 3 km suburban, 12 km rural. For specialty / destination retail (book stores, art supply, music shop), expand to 25+ km.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Product selection / breadth
- Staff knowledge + helpfulness
- Visual merchandising / store atmosphere
- Pricing transparency / value
- Cleanliness + organization
- Checkout speed
- Returns + customer service
- Local-maker / curation story

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **17** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Local-maker consignment program	Named regional makers + consignment terms	12 named local makers on consignment — 60/40 split — adds \$44k/yr revenue with \$0 inventory cost
Workshop / class series	Capacity + class topic demand	\$48 candle-making class — 12 students × 2 sessions/mo = \$13.8k/yr + builds email list

Loyalty + birthday club	POS-integrated loyalty (Square, Shopify, Lightspeed)	Birthday \$20 reward — 1,400 emails × 22% redeem = 308 visits × \$48 ATV = \$14.7k incremental
In-store + online unification (BOPIS)	Shopify POS unified inventory + pickup workflow	Buy-online-pickup-in-store — captures 28% incremental sales at \$0 fulfillment cost
Email + SMS list growth	Klaviyo benchmark open + conversion rates	Pop-up "join for 10% off" — captures 22% of foot traffic — 6,400 emails in year 1
Influencer micro-creator partnership	Named local IG / TikTok creators + audience fit	Named local lifestyle creator (28k IG) — comp \$200 outfit — drives 80 store visits in 2 weeks
Subscription / box program	Customer survey + repeat-purchase data	\$45/mo curated gift box — 120 subscribers = \$65k/yr predictable revenue
Window display + sandwich-board strategy	Foot-traffic count + window conversion	4-week rotation window — measured by door counts — best windows lift traffic 18%
Pop-up / shop-in-shop hosting	Complementary local brands seeking pop-up	Host named local jeweler for Dec — \$400/wk fee + 12% sales — adds \$3,800 net + cross-traffic
Photo / video for Instagram + TikTok	Content cadence benchmark + format	3 IG reels / wk — named local content creator — 2× organic reach in 90 days
Live shopping / Whatnot	Platform fit + product-selection economics	Sat 7pm live shopping — 80 viewers × \$32 ATV × 18% conv = \$460/show + new customer base
Trade-in / consignment circular	Customer demand + condition standards	Used-book trade-in — store credit at 25% — drives 14% of transactions, fills inventory at low cost
Charity tie-in fundraisers	Named local nonprofits + beneficiary mechanics	"Round up at register for [named local food bank]" — \$4,200/yr + community press
Private appointment shopping	Customer demand + staff capacity	After-hours private appointment — \$48 ATV becomes \$124 ATV — books 3-4/wk
Visual merch + store-staging upgrade	Named retail merch consultants + ROI benchmark	Quarterly merch refresh by named retail consultant — \$1,200/visit — sales lift 8% in measurement window
Local SEO + Google Business Profile dom	GBP completeness + photo cadence + review velocity	Weekly photo posts + monthly review-collection campaign — Google Maps "near me" rank up 6 spots
Door-counter analytics	Traffic vs conversion benchmark	Hexa door counter — \$480 — surfaces 28% of traffic walks-out without engagement, prompts staffing change

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named maker / creator must exist in Etsy / Faire / Google Places
- Every benchmark must reference NRF / Shopify / Square published data
- Every loyalty / POS recommendation must reference real software (Square, Shopify, Lightspeed, Toast)
- Every named festival / market must be currently active

6. Cost scale (L5 — per-sector currency bands)

Specialty retail spend is small-business hospitality scale:

1 — Under \$200	Email setup, partnership outreach, GBP refresh
2 — \$200 to \$2,000	Loyalty software setup, photo shoot, signage, door counter
3 — \$2,000 to \$20,000	POS upgrade, store refresh, paid ad campaign, workshop space
4 — \$20,000 to \$100,000	Full store renovation, e-commerce buildout, second location
5 — Over \$100,000	Major expansion, warehouse, multi-store buildout

7. Persona schema (L6)

Schema: B2C_consumer

Required persona fields:

- Name + age + life stage
- Visit frequency (weekly regular vs special occasion)
- Average ticket + impulse vs planned
- Acquisition channel (foot traffic, IG, friend referral, GBP)
- Photo / share behavior
- LTV (visits/yr × ATV)

8. Cycle model (L7 — replaces calendar seasons)

Specialty retail has a strong holiday + tourist + local-event cycle:

Q4 holiday (Nov-Dec)	40-60% of annual sales for gift retail	Maximize merch, extend hours, holiday email cadence
Q1 inventory clean (Jan-Feb)	Slow + clearance	End-of-season clearance, planning, store refresh
Spring (Mar-May)	Mother's Day, graduation, weddings	Gift-occasion programming + bridal
Summer (Jun-Aug)	Tourist + summer market season	Outdoor sidewalk sale, market presence
Back-to-school (Aug-Sep)	School / dorm gift season	Dorm decor, school supply, locker decor

Local festival / event windows	Crafts fair, art walk, parade		Special programming + vendor presence
Annual: holiday postmortem	Q1 review of holiday performance	Plan next year inventory + staff schedule	

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Inventory turn rate / dead stock — universal — frame as POS analytics + clearance discipline
- Staff turnover + knowledge gaps — universal — frame as training + tipped roles
- Window walk-by conversion — universal — frame as visual merch refresh
- Online pricing transparency wars — universal — frame as in-store experience differentiator
- Big-box / Amazon competition — universal — frame as curation + service moat
- Theft / shrink — universal — frame as POS controls + door analytics

10. Report sections (L9 — section list and order)

- 1 **Store snapshot** — Sq ft, ATV, monthly traffic, top categories, named comp set
- 2 **Top 5 differentiation moves** — Drawn from 17-category playbook
- 3 **Quick wins (under \$2k)** — Process + partnership wins
- 4 **Local maker + partnership map** — Named makers / influencers / pop-ups
- 5 **Customer personas (4 + lost shopper)** — Regular / gift-buyer / tourist / impulse
- 6 **Annual cycle plan** — 7-window calendar tied to holidays + local events
- 7 **Comp set comparison** — Named retailers + their gaps
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Avg ticket value (ATV)
- Foot traffic count vs conversion rate
- Sales per square foot (\$)
- Email + SMS list size + opt-in rate
- Inventory turn (target > 6× / yr for specialty)
- Loyalty member % of transactions

44-45.auto_dealer — Auto dealers (new + used cars, RVs, motorcycles, boats)

Business model: B2C_appointment | **Profile ID:** retail.auto_dealer

Covers NAICS: 441110, 441120, 441210, 441222, 441227

1. Data sources (replaces v3 hospitality stack)

DealerRater + Cars.com reviews	Dealership-specific reviews + ranking	Subscription	Google Reviews
Edmunds + Kelley Blue Book	Inventory + pricing intelligence	Free / sub	Cox AutoTrader
CDK / Reynolds DMS data	F&I; + service department metrics	Internal	Manual extract
NADA / state dealer assoc reports	Industry margin + benchmark data	Member-only	NADA Data
Manufacturer dashboard (CSI / SSI)	OEM customer satisfaction scores	Direct OEM	Internal
CarGurus / TrueCar	Listing performance + lead conversion	Subscription	Direct
Manheim / ADESA auction data	Used inventory acquisition pricing	Subscription	Local auction

2. Competitor radius

15 mi for new car (consumers shop 3-5 dealers), 25-50 mi for used (shop wider), 100 mi for specialty / luxury / RV / boat. Same-OEM competitors prioritized for new car.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Sales process pressure / honesty
- Trade-in offer fairness
- Financing transparency
- Service department quality
- Loaner / shuttle availability
- Communication during service
- Warranty / followup
- Online inventory accuracy

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

F&I; product penetration audit	Named F&I; providers benchmark attach rates	+	Add Resistance All / Zurich F&I; menu — VSC attach lifts from 32% to 48% — \$620 PRU gain
Service department customer experience	Named DMS service module + CSI program	+	CDK service-lane tablet check-in — DVI photos sent to customer — service ARO up \$48
Used inventory acquisition strategy	Manheim + private-party lease-return mix	+	In-house "we buy cars" campaign — sources 24 units/mo direct from owner — saves \$1,200/unit auction fee
Internet sales team + BDC	Lead-conversion benchmark + CRM (VinSolutions, ELEAD)	+	BDC team for first-touch — internet lead close rate from 8% to 18%
Online inventory + photo refresh	CarGurus + Cars.com listing performance benchmarks		Professional photography — 28 photos per unit + walk-around video — VDP views up 42%
Service contract + accessory attach	Pre-load accessory pricing + finance integration	+	Pre-load \$1,200 in accessories on used inventory — 80% retention into financed deal
Loaner fleet expansion	Service-write-up benchmark + repeat business	+	14-car loaner fleet (vs current 4) — service CSI lifts 12 points + repeat business 14%
EV-service capability + chargers	OEM EV training + named charger installer		4 ChargePoint stations — Wisconsin grant 50% — captures growing EV service market
Manufacturer recall + outreach campaign	NHTSA recall lookup + customer database		Outreach to 800 owners with open recalls — 40% redemption — adds \$96k service revenue
Reputation management program	DealerRater + Google Review velocity + response		Daily review monitoring + response — Google rating from 4.0 to 4.6 in 6 mo
Employee retention / pay-plan redesign	Tenure benchmark + comp structure	+	Tiered tenure bonus — sales staff 18-mo retention from 32% to 64%
Pre-purchase inspection niche	Used-car private market + inspection demand	+	PPI service for private buyers — \$189/inspection — 80/mo = \$15k/mo + lead-gen for trade-in
Fleet / commercial sales	Local contractor + government fleet demand		Named municipality fleet contract — 14 trucks/yr at fleet pricing
Ancillary trim / cosmetic services	Tint, ceramic coat, PPF demand		In-house tint + PPF — \$480 tint, \$1,800 PPF — 60% margin on \$4k/wk service revenue

5. Specificity rules (L4 — what 'real named entity' means here)

- Every F&I; product must reference real provider + product name
- Every DMS / CRM recommendation must reference real software
- Every benchmark must come from NADA / state dealer assoc / OEM CSI
- Every named partner / vendor must be a verifiable supplier

6. Cost scale (L5 — per-sector currency bands)

Auto dealers operate at large-format retail capex bands:

1 — Under \$1,000	Photo shoot, online listing refresh, training session
2 — \$1,000 to \$10,000	BDC software, F&I; menu, signage
3 — \$10,000 to \$100,000	DMS module add, service tablet, EV charger install
4 — \$100,000 to \$1M	Showroom remodel, body shop equipment, service bay add
5 — Over \$1M	New facility, second franchise, full renovation

7. Persona schema (L6)

Schema: B2C_consumer (with extended F&I; + service lifecycle fields)

Required persona fields:

- Name + life stage + financing profile
- Vehicle replacement cycle + brand loyalty
- Service preference (dealer vs independent)
- Trade-in vs cash-out preference
- Online research depth + visit count before close
- Lifetime value (vehicles purchased + service spend over 10 yrs)

8. Cycle model (L7 — replaces calendar seasons)

Auto retail runs on model year + tax + season cycles:

Model-year-end (Jul-Sep)	New-model launches + outgoing-model clearance	Aggressive incentive marketing on outgoing inventory
Tax-refund (Feb-Apr)	Down-payment cash availability	Used-car promo + first-time-buyer programs
Year-end (Nov-Dec)	Sales push + business-tax-year buyers	Sign + Drive promos + commercial fleet sales
Spring (Mar-May)	Convertible / motorcycle / boat season	Specialty inventory promo + outdoor display
Service: warranty windows	Manufacturer maintenance schedule	Outreach campaign at intervals
Service: tire / brake season	Winter prep + summer prep	Tire promo Oct-Nov, brake/AC Apr-May
Annual: OEM CSI cycle	Manufacturer customer satisfaction	Internal CSI improvement + QBR with rep

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Pressure / sleazy sales reputation — universal — frame as no-haggle / transparent process
- F&I; aggressive add-ons — universal — frame as menu-presentation reform
- Service repair surprises — universal — frame as DVI photo + estimate transparency
- Online inventory accuracy — universal — frame as DMS sync + photo refresh
- Loaner unavailability — universal — frame as fleet expansion ROI
- Long service write-up wait — universal — frame as service-lane tablet + appointment
- Trade-in lowball perception — universal — frame as KBB + Black Book transparency

10. Report sections (L9 — section list and order)

- 1 **Dealership snapshot** — Brand, units / mo, ASP, F&I; PRU, service ARO, named comp set
- 2 **Top 5 strategic moves** — Drawn from 14-category playbook
- 3 **F&I; + service deep-dive** — Named providers + attach + ARO benchmarks
- 4 **Customer personas (5 + lost shopper)** — New-car / used-car / service / fleet / first-time
- 5 **Inventory + acquisition strategy** — New + used mix + sourcing channels
- 6 **Reputation + CSI improvement plan** — Named platforms + tactic-by-platform
- 7 **Annual cycle plan** — Model + tax + season calendar
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Units sold per month (new + used)
- Front-end gross + F&I; PRU (\$)
- Service ARO (avg repair order \$)
- CSI score from OEM
- Internet lead close rate (target > 15%)
- Service / sales attach rate (target > 60% of buyers using service)

44-45.ecommerce_dtc — E-commerce / DTC consumer brands (online-only or omnichannel)

Business model: B2C_online | **Profile ID:** retail.ecommerce_dtc

Covers NAICS: 454110 (electronic shopping), 459xxx (online-only specialty)

Modern formats (Part 2): DTC Brand, Subscription Box, Print-on-demand Shop, Amazon FBA Brand

1. Data sources (replaces v3 hospitality stack)

Shopify / WooCommerce admin	Conversion + AOV + repeat-rate data	Internal	Direct export
Klaviyo / Mailchimp	Email + SMS performance	Internal	CSV export
Meta Ads + Google Ads	CAC + ROAS + creative perf	Internal	Direct API
SimilarWeb	Competitor traffic + source mix	Subscription	Manual
Amazon Seller Central	Marketplace + buy-box + reviews	Internal	Helium 10
Trustpilot + Google Shopping reviews	Brand reputation	Free	Manual
G2 / Capterra (B2B SaaS DTC)	B2B-buyer reviews	Free	Manual
Trade publications + IR data	Industry trend + benchmark	Free / sub	Direct

2. Competitor radius

No physical radius — competitors are defined by category + price band + audience overlap. Use SimilarWeb audience overlap metric.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Product quality vs marketing claims
- Shipping speed + cost transparency
- Returns + customer service experience
- Site + checkout UX
- Brand authenticity / story
- Subscription / billing transparency
- Photo + product accuracy
- Sustainability / ethics narrative

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **16** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Email lifecycle automation buildout	Klaviyo benchmark flows + revenue / send	Klaviyo welcome + abandoned cart + post-purchase + winback — adds 22% of total revenue from email
SMS list growth + automation	Postscript / Attentive benchmarks	Postscript SMS — 18% of email-revenue at 2× conversion — adds 8% of revenue at low cost
Subscription / membership tier	Recharge / Bold benchmarks + churn data	Auto-replenish at 12% discount — 14% of customers convert — 4× LTV vs one-time
Influencer / creator gifting program	Aspire / Grin / GRIN benchmarks	40 micro-creator gifts/mo — drives \$14k attributed revenue at \$80 CAC
Paid social creative testing	Meta + TikTok creative testing methodology	4 new creatives / wk — winners scaled — Meta CPA from \$48 to \$32 in 6 wks
Affiliate / referral program	Refersion / GoAffPro setup + commission structure	Affiliate program — 12% commission — 40 active affiliates drive 8% of revenue
Product reviews + UGC engine	Yotpo / Okendo / Stamped benchmarks	Yotpo review request flow — 22% review-rate, drives 14% conversion lift on PDPs
Site UX + conversion audit	Hotjar / FullStory + tested heuristics	Checkout simplification — guest checkout, address autofill — conversion lifts from 2.4% to 3.1%
Wholesale / Faire / retailer expansion	Faire + Tundra + retailer outreach	60 retailers via Faire — adds \$14k/mo wholesale at 50% margin
Amazon channel strategy (or exit)	Helium 10 + buy-box + ads benchmarks	Amazon hybrid — direct + FBA — adds 28% revenue at 22% margin (vs 48% DTC)
TikTok Shop + live shopping	TikTok Shop policy + creator partnership	TikTok Shop launch + 4 creator collabs — \$28k in 60 days at \$14 CAC
CRO / on-site personalization	Dynamic Yield / Klaviyo personalization	Personalized PDP recommendations — AOV up \$12 — 8% revenue lift
PR + earned media campaign	Named PR firms + outlet fit	Named PR firm 6-mo retainer — 4 placements (Forbes, BuzzFeed) — drives 2× brand search
Bundling + cross-sell strategy	Bundle.app + ReConvert + benchmarks	Post-purchase upsell offer — 12% conversion at \$24 AOV add
Retention email segmentation	RFM segmentation + per-segment offer	VIP segment (top 10% LTV) gets early access + 20% off — drives 28% repeat rate vs 12% average

Sustainability / impact narrative	1% for the Planet, B Corp, named cert	B Corp certification — adds 14% younger-cohort conversion + media mentions

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named platform must be real software (Klaviyo, Postscript, Yotpo, Recharge)
- Every benchmark must come from named industry source (Klaviyo benchmarks, Shopify Plus, Triple Whale)
- Every named PR / agency partner must be a real firm
- Every named retailer / marketplace must be currently operating

6. Cost scale (L5 — per-sector currency bands)

DTC e-commerce spend bands are software + ads + creative scale:

1 — Under \$200	App install, email setup, creative iteration
2 — \$200 to \$2,000	PR pitch, paid creative test, influencer gifts
3 — \$2,000 to \$20,000	CRO retainer, PR firm month, full lifecycle build
4 — \$20,000 to \$100,000	Site rebuild, packaging refresh, paid scale
5 — Over \$100,000	New product line, warehousing, omnichannel buildout

7. Persona schema (L6)

Schema: B2C_consumer (online-first, digital-touchpoint focused)

Required persona fields:

- Name + age + digital behavior
- Acquisition channel (paid social, organic, referral, search)
- AOV + frequency + return-rate
- Loyalty / subscription propensity
- Reviews + content-creation behavior
- LTV (orders × AOV × retention years)

8. Cycle model (L7 — replaces calendar seasons)

E-commerce runs on weekly + holiday + paid-media cycles, no physical season:

Q4 holiday (Nov-Dec)		40-60% of annual revenue	BFCM prep starts Sep, max creative + email cadence
Q1 winback (Jan-Feb)		Slow + new-year resolution	Sale + winback campaigns + retention focus

Spring (Mar-May)	Mother's Day + Memorial Day	Gift-occasion campaigns + new-collection launches
Summer (Jun-Aug)	Travel + outdoor demand	Seasonal promos + summer campaigns
Back-to-school (Aug-Sep)	Demand for school / dorm products	Targeted creative + bundles
Daily / weekly: paid + email cadence	Always-on revenue engine	Creative testing + email send schedule
Annual: pricing + cost review	Margin + competitive review	Cost increases + price + COGS audit

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Shipping cost / speed expectations vs Amazon — universal — frame as transparency + threshold
- Returns + reverse logistics cost — universal — frame as policy + flat-fee return
- Subscription billing surprise / churn — universal — frame as transparent cancel + skip
- Customer service response time — universal — frame as Gorgias / Zendesk SLA
- Product accuracy / photos vs reality — universal — frame as photography + size guides
- Email frequency complaints — universal — frame as preference center
- Paid CAC inflation — universal — frame as creative diversification + organic

10. Report sections (L9 — section list and order)

- 1 **Brand snapshot** — Revenue, AOV, repeat-rate, channel mix, named comp set
- 2 **Top 5 growth moves** — Drawn from 16-category playbook
- 3 **Lifecycle + retention plan** — Named flows + segmentation strategy
- 4 **Channel-mix optimization** — Paid + organic + wholesale + marketplace
- 5 **Customer personas (4 + lost customer)** — New / repeat / VIP / churned
- 6 **CRO + UX audit findings** — Top friction points + estimated lift
- 7 **Competitor channel-mix comparison** — SimilarWeb-derived comp data
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Avg order value (AOV)
- Customer acquisition cost (CAC) by channel
- Repeat-purchase rate within 90 days
- LTV : CAC ratio (target > 3:1)
- Email + SMS revenue % of total (target > 25%)
- Site conversion rate (target > 2.5% e-commerce)

Engineering note for this sector

Sector 44-45 is where v3 mostly works for the brick-and-mortar sub-cluster (just needs profile-format refactor) but needs significant new logic for auto dealers and pure DTC e-commerce. Build retail.specialty_brick_mortar first (fits Phase 1), then retail.auto_dealer (Phase 2), then retail.ecommerce_dtc (Phase 5 — it's a fundamentally different beast).

CHAPTER 62 — HEALTH CARE & SOCIAL ASSISTANCE

Private medical practices, dental, mental health, other practitioners, child care, residential care. The biggest B2C-appointment opportunity outside hospitality.

Covers 39 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.

The v3 problem in this sector

A dental practice has reviews on HealthGrades + ZocDoc, not TripAdvisor. Patients are decision-makers but insurance is the silent gatekeeper. v3 has zero concept of insurance networks, payer mix, or patient-acquisition funnels. Four distinct profiles required.

Sub-clusters in this sector

This sector splits into 4 sub-clusters because the business models inside diverge enough that one profile cannot serve them all. Each sub-cluster gets its own complete spec below.

healthcare.medical_p ractice	Private medical practices (primary care, specialists, urgent care)	B2C_appointment	621111, 621112, 621210 (note: dental separately), 621399, 621498
healthcare.dental_pra ctice	Dental practices (general, pediatric, ortho, oral surgery, periodontics)	B2C_appointment	621210
healthcare.mental_he alth	Mental health practices (therapists, psychologists, psychiatrists, group practices)	B2C_appointment	621330, 621112 (psychiatrists), 621399, 621498
healthcare.child_care	Child day care, preschools, after-school programs	B2C_appointment	624410, 611110 (preschool component), 624110

62.medical_practice — Private medical practices (primary care, specialists, urgent care)

Business model: B2C_appointment | **Profile ID:** healthcare.medical_practice

Covers NAICS: 621111, 621112, 621210 (note: dental separately), 621399, 621498

Modern formats (Part 2): Concierge / DPC Practice, Direct-Pay Med Spa, Wellness Clinic, Functional Med Practice, GLP-1 / Weight-Loss Clinic, Hormone Clinic

1. Data sources (replaces v3 hospitality stack)

HealthGrades + Vitals	Doctor reviews + ratings + specialty	Free / sub	Direct
ZocDoc	Booking + review + insurance accepted	Subscription	Direct
Google Places + Reviews	Practice-level reviews	\$0.02	Yelp
CMS / Medicare data	Procedure volumes + payments by NPI	Free public	Direct
NPI registry	Provider name + specialty + address	Free	Direct
State medical board	License + complaint history	Free	Direct office
Insurance carrier directories	In-network status by plan	Free	Direct call
Local employer benefits directory	Named local employers + their carriers	Free / direct	Chamber

2. Competitor radius

5 mi urban, 15 mi suburban, 35 mi rural. Specialists draw 50+ mi. Concierge / cash-pay practices regional (75+ mi). Insurance + referral patterns matter more than radius for traditional practices.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Wait time vs appointment
- Doctor listening + bedside manner
- Office staff friendliness
- Insurance billing accuracy
- Appointment availability
- Office cleanliness + comfort
- Telehealth experience
- Communication after appointment
- Prescription / refill responsiveness

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **15** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Online booking + patient portal	Athenahealth / Phreesia / Mend benchmarks	Phreesia online booking — captures 38% of appointments after-hours, fills 4-day-out gaps
Telehealth / hybrid expansion	Doxy.me / SimplePractice / Zoom for Healthcare	Telehealth for follow-up + simple visits — captures 22% visit volume at \$0 facility cost

Concierge / DPC pivot	DPC pricing model + patient panel size	Hybrid DPC — \$79/mo for 200 patients = \$190k recurring + capacity for fewer high-touch patients
Cash-pay + transparent pricing	Procedure-level benchmark + pricing partnership	Sesame partnership — published \$89 office visit + \$145 lab panel — captures uninsured + deductible patients
GLP-1 / weight-loss / hormone clinic add-on	Compounded GLP-1 + named telehealth platforms	Weight-loss clinic add-on — \$399/mo membership × 80 patients = \$32k/mo recurring
Aesthetic / Botox / IV add-on	Aesthetic add-on margins + training	Allergan partner program — Botox + filler — \$1,200 avg ticket × 4 patients/wk = \$20k/mo
Lab / imaging in-office	CLIA waived tests + named lab partners	In-office LabCorp / Quest patient service center — adds \$4k/mo + improves patient satisfaction
Insurance-network expansion / negotiation	Local employer carriers + carrier contracting	Add named local United HMO contract — adds 800 covered lives × 15% utilize = 120 new patients
Saturday / evening hours	Working-population demand + staff cost	2 Saturday mornings/mo — drives 24 appointments/Sat × \$148 reimburse = \$7k/mo
Reputation management review velocity	Birdeye / Solv / Press Ganey + review request automation	Birdeye SMS post-visit review — Google rating 4.2 → 4.7 in 6 mo
Spanish-speaking marketing	Census Hispanic % + named Spanish-language outlets	Bilingual Spanish-speaking front desk + website — captures Spanish-speaking population (huge gap most markets)
Care-coordination + chronic care mgmt	CMS CCM billing codes + workflow	CCM billing — \$42/mo × 200 patients with 2+ chronic conditions = \$8.4k/mo new revenue
Specialty referral relationships	Named specialists + referral coordination	Coordinated referrals to 4 named specialists — closes the loop, builds reputation
Workplace / corporate clinic contract	Named local employers + on-site model	Named 400-employee employer — onsite clinic 2 days/wk — \$14k/mo recurring
Patient-experience training	AAFP / specialty-society + Press Ganey training	Patient-experience training for staff — Press Ganey scores +18 percentile

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named platform must be real (Athenahealth, eClinicalWorks, Phreesia, Doxy.me, Sesame, Solv)
- Every billing code / program must be real (CCM, CPT codes, Medicare programs)
- Every named insurer must be real + currently operating in this market
- Every named partnership must be a verifiable employer / lab / specialist

- Every benchmark must come from MGMA / AAFP / specialty society

6. Cost scale (L5 — per-sector currency bands)

Medical practice spend bands are professional-services + medical-equipment scale:

1 — Under \$500	Software trial, partnership outreach, contract negotiation
2 — \$500 to \$5,000	Booking software, signage, training, marketing campaign
3 — \$5,000 to \$50,000	New service line equipment, EHR module, lab partnership
4 — \$50,000 to \$500,000	Major equipment, second location, EHR migration
5 — Over \$500,000	New facility, surgery center add, large practice acquisition

7. Persona schema (L6)

Schema: patient (insurance + decision-driver model)

Required persona fields:

- Name + age + condition profile
- Insurance carrier + plan + deductible status
- Acquisition channel (insurance directory, referral, search, employer)
- Visit cadence (annual physical vs chronic care vs episodic)
- Telehealth willingness
- Cash-pay willingness for non-covered services
- LTV (visits/yr × payer mix-weighted reimbursement)

Example persona: Sarah, 44, working mother, employer-sponsored UnitedHealthcare PPO

Schedules around work + kids. Acquisition: searched UHC directory + Google reviews. Spend trigger: same-week availability for sick visits. 5-star trigger: doctor listened to her concerns + sent prescription before she left. Telehealth willing for follow-ups. Cash-pay for cosmetic only. LTV: 4 visits/yr × \$148 weighted reimburse + Botox add-on (\$1,200/yr) = \$1,792/yr × 8 yrs = \$14k.

8. Cycle model (L7 — replaces calendar seasons)

Medical practices run on insurance + flu + back-to-school + tax-deductible cycles:

Open enrollment (Oct-Dec)	New plan year decisions + HSA spend	Pre-Jan booking surge for elective + checkups	
New year (Jan-Mar)	Deductible reset — patients delay care	Cash-pay specials + financing options	
Spring (Apr-May)	Back-to-routine physicals + sports	Sports physical + annual checkup push	

Back-to-school (Aug-Sep)	Physicals + immunizations surge	Capacity surge + Saturday hours
Flu season (Oct-Feb)	Sick visit volume peak	Walk-in / urgent capacity + Saturday
Year-end (Nov-Dec)	HSA spend-down + deductible-met patients	Promote HSA-eligible care + cosmetic
Quarterly: insurance contract review	Carrier negotiations + new contracts	Review payer mix + negotiate higher reimbursement

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Wait time / appointment availability — universal — frame as scheduling + telehealth
- Insurance billing surprises — universal — frame as upfront benefit verification
- Hold-time / phone access — universal — frame as patient portal + texting
- Office staff coldness — universal — frame as patient-experience training
- Doctor not listening — universal — frame as visit-time discipline + advance form
- Refill / prescription delays — universal — frame as portal + automated workflows
- Telehealth tech friction — universal — frame as platform UX

10. Report sections (L9 — section list and order)

- 1 **Practice snapshot** — Specialty, providers, payer mix, monthly visits, named comp set
- 2 **Top 5 growth + experience moves** — Drawn from 15-category playbook
- 3 **Service-line expansion plan** — Named cash-pay + add-on opportunities
- 4 **Insurance + payer-mix strategy** — Named carriers + contract opportunities
- 5 **Patient personas (4 + lost patient)** — Wellness / chronic / episodic / cash-pay
- 6 **Reputation + comm strategy** — Named platforms (HealthGrades, ZocDoc, Google)
- 7 **Annual cycle plan** — Insurance + clinical calendar
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Monthly visits + new-patient %
- No-show rate (target < 8%)
- Avg reimbursement per visit (payer-weighted)
- Cash-pay revenue % of total (target growing > 15%)
- Patient retention 12-mo
- Online review rating + velocity (HealthGrades, ZocDoc, Google)

62.dental_practice — Dental practices (general, pediatric, ortho, oral surgery, periodontics)

Business model: B2C_appointment | **Profile ID:** healthcare.dental_practice

Covers NAICS: 621210

Modern formats (Part 2): Membership Plan Practice, Sleep Apnea Specialist, Sedation Dentistry, Cosmetic-Only Practice

1. Data sources (replaces v3 hospitality stack)

Google Reviews	Places	+	Practice-level reviews	\$0.02	Yelp
HealthGrades		+	Dentist-specific reviews	Free	Direct
RateMDs					
ZocDoc Direct	+	Smile	Booking + insurance accepted	Subscription	Direct
ADA directory		member	Industry benchmarks	Member-only	Public summary
DentalIntel	/	Dental	Practice analytics + benchmarks	Subscription	Direct
Whale benchmarks					
State dental board			License + complaint history	Free	Direct
Insurance dental directories		carrier	In-network status	Free	Direct

2. Competitor radius

5 mi urban, 12 mi suburban, 25 mi rural. Specialty (ortho, oral surgery, sleep) draws 50+ mi. Cosmetic-only practices regional.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Pain management + comfort
- Wait time vs appointment
- Insurance billing accuracy
- Sedation + anxiety management
- Cosmetic / aesthetic skill
- Cleanliness + sanitation
- Front-desk friendliness
- Treatment plan transparency

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

In-house membership plan (uninsured)	Kleer / Plan Forward / DentalHQ structures	Kleer in-house plan — \$35/mo × 280 members = \$118k/yr recurring + 2× cleanings + 1.4× services attach
Sedation / anxiety services	IV sedation + nitrous + oral sedation training	IV sedation training + license — captures 22% of fearful patients others won't treat
Cosmetic / Invisalign add-on	Invisalign / Spark / SureSmile partner programs	Invisalign + named lab partner — \$4,800 avg case × 30 cases/yr = \$144k incremental
Same-day crowns (CEREC)	CEREC equipment + benchmark utilization	CEREC same-day crown — \$65k equipment, 4 crowns/wk × \$1,400 = \$5,600/wk = \$291k/yr revenue
Implant + bone-graft service-line	Implant training + named OEM partner	Add implant placement (vs refer-out) — \$2,400 placement + \$1,800 crown = \$1.2M/yr at 60% margin
Pediatric niche / family practice	Local family demographic + child-friendly facility	Pediatric-friendly office — captures multi-generation family base, 60% of new from referral
Online booking + patient portal	Dentrix / Eaglesoft / Solutionreach integrations	Solutionreach online booking — captures 32% of bookings after-hours
Reputation management review velocity	Birdeye / RevenueWell + automated review request	RevenueWell post-appointment SMS — Google rating 4.0 → 4.6 + 4× review velocity
Sleep apnea / oral appliance	Named DME + medical billing partner	Sleep apnea oral appliance — \$2,200 device + medical insurance billing — 6 cases/mo = \$13k/mo
Insurance contract optimization	Carrier fee schedules + drop-in-network analysis	Drop lowest-paying PPO (Delta), increase fee schedule on remaining — \$84k/yr fee increase capture
Spanish-speaking marketing	Bilingual staff + community outreach	Bilingual front desk + Spanish website — captures growing market segment
Recall / re-care system	Open Dental + Lighthouse 360 recall systems	Lighthouse 360 automated recall — re-care rate from 38% to 76% — adds \$144k/yr predictable revenue
Lobby treatment-room amenities	Patient comfort upgrades + cost	Ceiling-mounted TVs, blankets, headphones — review sentiment +18, anxiety reviews drop
New-patient welcome + financing	CareCredit / LendingClub partnerships	CareCredit financing offered upfront — accepts 6-mo no-interest — case acceptance from 48% to 72%

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named software must be real (Dentrix, Eaglesoft, Open Dental, Lighthouse 360, RevenueWell, Solutionreach)
- Every named partner program must be real (Invisalign, Kleer, CareCredit, Patterson Dental)
- Every billing code / program must be real (CDT codes, in-network agreements)
- Every benchmark must come from ADA / AGD / DentalTown / DentalIntel

6. Cost scale (L5 — per-sector currency bands)

Dental practice bands are professional-services + medical-equipment scale:

1 — Under \$500	Plan setup, partner outreach, training session
2 — \$500 to \$5,000	Software upgrade, signage, marketing campaign, comfort items
3 — \$5,000 to \$50,000	New equipment line, training course, op rebuild
4 — \$50,000 to \$500,000	CEREC + 3D imaging + laser + implant equipment
5 — Over \$500,000	New facility, second location, full op rebuild

7. Persona schema (L6)

Schema: patient (with dental-specific lifecycle)

Required persona fields:

- Name + age + dental anxiety profile
- Insurance vs membership vs cash-pay
- Service preference (preventive only vs cosmetic vs full)
- Family attached (other family members at practice)
- Acquisition channel (insurance directory, referral, search, employer)
- LTV (visits/yr × case acceptance × case value × loyalty years)

8. Cycle model (L7 — replaces calendar seasons)

Dental practices run on insurance + school + holiday cycles:

Open enrollment (Oct-Dec)	Insurance + HSA spend deadline	Pre-Jan checkup + cosmetic surge
Use-it-or-lose-it (Nov-Dec)	Annual benefit max approaching	Patient outreach to use remaining benefits
New year deductible reset (Jan-Mar)	Patients delay non-urgent care	Membership plan + financing emphasis
Wedding / event prep (Mar-May)	Cosmetic + whitening surge	Whitening + Invisalign promo

Back-to-school (Aug-Sep)		Pediatric + sealant + ortho surge	School physical + sealant + ortho consult
Daily: discipline	schedule	Hygienist + doc bay + treatment plan flow	Schedule discipline + same-day treatment
Quarterly: recall	re-care	Patient retention via recall	Automated recall + missed-recall outreach

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Pain / anxiety experience — universal — frame as sedation + comfort
- Insurance + billing surprises — universal — frame as upfront treatment plan + estimate
- Wait time vs appointment — universal — frame as schedule discipline
- Cost shock for major treatment — universal — frame as financing + phased treatment
- Front-desk attitude — universal — frame as patient-experience training
- Recall / re-care drop — universal — frame as automated system
- Hygienist consistency — universal — frame as standardized protocols

10. Report sections (L9 — section list and order)

- 1 **Practice snapshot** — Doctors, hygienists, ops, monthly production, payer mix, named comp set
- 2 **Top 5 growth + retention moves** — Drawn from 14-category playbook
- 3 **Service-line expansion plan** — Named opportunities by margin + capacity
- 4 **Membership / insurance / cash-pay strategy** — Payer-mix optimization plan
- 5 **Patient personas (4 + lost patient)** — Family / cosmetic / fearful / cash-pay
- 6 **Recall + retention engine** — System + automation + outreach plan
- 7 **Annual cycle plan** — Insurance + school + cosmetic calendar
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Monthly production + collection
- Case acceptance rate (target > 65%)
- Hygiene re-care rate (target > 75%)
- New-patient count + cost-per-acquisition
- Avg revenue per patient
- Membership plan enrollment count

62.mental_health — Mental health practices (therapists, psychologists, psychiatrists, group practices)

Business model: B2C_appointment | **Profile ID:** healthcare.mental_health

Covers NAICS: 621330, 621112 (psychiatrists), 621399, 621498

Modern formats (Part 2): Online-Only Therapy Practice, Group Therapy Specialty, Esketamine Clinic, EMDR Specialty Practice

1. Data sources (replaces v3 hospitality stack)

Psychology directory	Today	Therapist listings + insurance + specialties	Subscription	Direct
TherapyDen	+	Niche-aligned therapist directories	Free / sub	Direct
Inclusive Therapists				
Google HealthGrades reviews	+	Practice-level reviews	\$0.02	Yelp
SimplePractice	/	Practice management benchmarks	Subscription	Direct
TherapyNotes	/			
TheraNest				
Insurance behavioral-health directories		In-network status	Free	Direct
EAP (Employee Assistance) panels		Named EAP networks (ComPsych, CareBridge, Lyra)	Direct application	EAP listing
State licensing board		License + complaint history	Free	Direct

2. Competitor radius

Online-only — defined by state licensure (telehealth across state lines requires licensure). In-person: 8 mi urban, 20 mi suburban, 50 mi rural. Specialty (eating disorder, OCD, trauma) draws 60+ mi.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Therapist match / rapport
- Booking + cancellation flexibility
- Insurance + sliding-scale options
- Telehealth experience
- Office comfort + privacy
- Communication + responsiveness

- Specialty fit
- Wait for first appointment

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **12** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Niche specialization positioning	Underserved demographic + specialty demand	LGBTQ+ specialization + Inclusive Therapists profile — fills practice in 6 mo + drives premium \$/session
Group therapy add-on	Group format + reimbursement code (90853)	DBT group — 8 members × \$65/session × 24 sessions = \$12.5k revenue at lower per-clinician hour
Online + telehealth scale	SimplePractice / Doxy.me + multi-state licensure	Telehealth-only model — eliminates office overhead — \$0 facility cost, full schedule fills
Insurance vs cash-pay strategy	Local payer rates + cash-pay benchmark	Drop lowest-reimbursing carrier (BCBS at \$80) — replace with cash-pay at \$185 — increases hourly rate 130%
EAP panel expansion	Named EAP networks (Lyra, ComPsych, CareBridge, Modern Health, Spring)	Lyra panel — \$135-180/session — fills schedule + steady referral pipeline
Couples / family therapy specialty	Gottman + EFT training + market demand	Gottman Level-3 couples therapy — \$245/session vs \$185 individual — 8 sessions/wk premium
Online directory profile optimization	Psychology Today + TherapyDen + Inclusive optimization	Optimized PT profile — 40+ profile views/mo → 4-6 inquiries → 2-3 new clients
Workshop / group educational program	Topic demand + paid group format	8-week anxiety skills group — \$400/person × 12 people × 4 cohorts = \$19.2k zero-individual-time revenue
Esketamine / IV ketamine partnership	Named clinics + medical-collaboration model	Refer-out partnership with named ketamine clinic — captures referrals back for talk therapy
Group practice expansion / W2 hires	Solo vs group economics + supervision model	Hire 2 W2 therapists at 60/40 split — adds \$200k/yr revenue at 40% margin to practice owner
Niche assessment offering	ADHD, autism, neuropsych assessment demand	Comprehensive ADHD assessment — \$1,800 self-pay × 8/mo = \$14k/mo (no insurance hassle)
Postpartum / perinatal niche	Postpartum Support International + named OB partners	PSI cert + named OB partnership — captures perinatal mood referrals (huge gap most markets)

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named directory must be real (Psychology Today, TherapyDen, Inclusive Therapists, ZocDoc)
- Every EAP must be a real network (Lyra, ComPsych, Modern Health, Spring Health, Magellan)
- Every named cert / training must be real (Gottman, EMDR, DBT, EFT, IFS)
- Every named software must be real (SimplePractice, TherapyNotes, TheraNest)
- Every billing code must be real (90791, 90834, 90837, 90847, 90853)

6. Cost scale (L5 — per-sector currency bands)

Mental health practice bands are professional-services + low-equipment scale:

1 — Under \$500	Directory profile, EAP application, training session
2 — \$500 to \$5,000	Software upgrade, marketing, advanced training, virtual setup
3 — \$5,000 to \$50,000	Office buildout, new clinician hire, group room setup
4 — \$50,000 to \$500,000	Practice expansion, second office, multi-clinician group
5 — Over \$500,000	Multi-location group practice acquisition or buildout

7. Persona schema (L6)

Schema: patient (mental health-specific privacy + match fields)

Required persona fields:

- Name + age + presenting concern
- Therapist match preference (gender, age, modality, demographic)
- Insurance vs cash-pay vs sliding-scale
- Telehealth vs in-person preference
- Frequency expectation (weekly vs bi-weekly vs as-needed)
- LTV (sessions/yr × rate × loyalty years — typical 30-60 sessions over 1-3 yrs)

8. Cycle model (L7 — replaces calendar seasons)

Mental health runs on academic + new-year + crisis cycles:

New year (Jan-Feb)	Resolution + therapy starts	Capacity surge + new-client onboarding
Spring (Mar-May)	Steady demand + couples	Couples / specialty programs
Summer (Jun-Aug)	College students home + family	Telehealth flexibility for travel
Back-to-school (Aug-Sep)	College student transition + parent stress	Capacity surge + young-adult focus
Holiday season (Nov-Dec)	Stress + family conflict surge	Maximum availability + extended hours
Post-holiday (Jan)	Major depression / seasonal affect	Capacity for crisis + acute cases

Quarterly: insurance + EAP review	Panel reimbursement + access	Drop low-paying panels + add new

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Wait for first appointment — universal — frame as group + telehealth + waitlist
- Insurance billing complexity — universal — frame as cash-pay + superbill + EAP
- Therapist match miss — universal — frame as consultation call before commitment
- Cancellation / no-show frequency — universal — frame as policy + late-fee discipline
- Privacy / confidentiality concerns — universal — frame as HIPAA + telehealth platform
- Insurance authorizations / utilization review — universal — frame as cash-pay positioning
- Burnout from full caseload — universal — frame as group + boundary protocols

10. Report sections (L9 — section list and order)

- 1 **Practice snapshot** — Clinicians, modalities, payer mix, monthly sessions, named comp set
- 2 **Top 5 growth + retention moves** — Drawn from 12-category playbook
- 3 **Niche specialization plan** — Underserved markets + training path
- 4 **Insurance + EAP + cash-pay strategy** — Payer-mix optimization plan
- 5 **Patient personas (4 + lost client)** — Insurance / cash-pay / niche / couples
- 6 **Directory + reputation strategy** — Psychology Today + TherapyDen optimization
- 7 **Annual cycle plan** — Academic + new-year + holiday calendar
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Active clients + caseload utilization
- Avg session rate (\$) — payer-weighted
- No-show rate (target < 8%)
- New-client conversion from inquiry (target > 60%)
- Avg client retention (sessions before discharge)
- Directory inquiry volume by source

62.child_care — Child day care, preschools, after-school programs

Business model: B2C_appointment | **Profile ID:** healthcare.child_care

Covers NAICS: 624410, 611110 (preschool component), 624110

Modern formats (Part 2): Montessori School, Reggio School, Forest School, Outdoor Preschool, Spanish-Immersion Preschool

1. Data sources (replaces v3 hospitality stack)

Google Places + Reviews	Reviews, photos, hours	\$0.02	Yelp	
Care.com + Winnie	Center reviews + booking + photo	Subscription	Direct	
NAEYC accreditation directory	Accredited centers + benchmarks	Free	Direct	
State child care licensing database	License + violation history	Free	Direct	
Census household + birth-rate data	Target demographic by ZIP	Free	Direct	
Local employer benefits directory	Named employers offering child care benefits	Free / direct	Chamber	
CCRC / state subsidy program	Subsidy acceptance + reimbursement	Free	Direct	

2. Competitor radius

5 mi for centers (parents minimize commute), 12 mi for specialty (Montessori, immersion, forest school). Daycare competes by drop-off-route geography more than radius.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Teacher quality + tenure
- Curriculum + activities
- Cleanliness + safety
- Communication with parents
- Pricing transparency
- Outdoor + play space
- Meal program
- Schedule flexibility
- Camera / app access

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Curriculum branding (Montessori, Reggio, Waldorf)	AMS / NAMTA / NAEYC standards + premium tuition	AMS Montessori certification — \$4k cert + teacher training — premium tuition \$325/wk vs \$245 baseline
NAEYC accreditation	NAEYC standards + 18-month process	NAEYC accreditation — drives 22% premium pricing + state subsidy preference
Camera + parent app program	Brightwheel / Procure / HiMama	Brightwheel daily updates + photos + invoicing — driving feature for new families, +18% conversion
Subsidy / Head Start participation	State subsidy rates + accreditation requirements	WI Shares subsidy participation — fills 12% empty slots at 85% of private rate = recurring fill
Corporate / employer partnership	Named local employers + tuition-benefit programs	Named employer (300 employees) — corporate partnership — \$480/family/yr employer subsidy + priority enrollment
After-school + summer camp programs	School schedule + capacity + ASP	After-school 3-6pm — \$185/wk × 14 kids = \$2,590/wk + summer camp 8wk × \$295/wk × 24 = \$56k
Sibling discount + family loyalty	Family-retention economics	15% sibling discount — captures multi-child family LTV (5-7 yr at center per family)
Specialty programming (music, Spanish, gym)	Adjunct provider partnerships + premium pricing	Add Spanish + music + gymnastics programs — \$35/wk per program × 22 enrolled × 3 = \$2,310/wk premium
Late pickup / drop-off flex	Working-parent demand + staffing	Extended hours 6:30am-6:30pm — \$20/wk premium — captures dual-earner working families
Sick-day backup care contract	Corporate benefits + named providers (Bright Horizons backup)	Backup care contract with named employer — \$148/day per child × 4-6 days/wk = \$4k/wk recurring
Outdoor / nature curriculum	Forest school certification + outdoor space	Outdoor / nature curriculum — premium tuition + waitlist — 3-6 mo waitlist drives FOMO + retention
Teacher retention + pay program	Local market wage + benefits + tenure data	\$2/hr above local benchmark + benefits — teacher tenure 1.2yr → 4.1yr = parent-perceived stability
Open house / discovery event	Tour conversion + scheduling	Bi-monthly Saturday open house — captures 18 family tours/mo → 4-6 enrollments
Pricing tier + waitlist strategy	Capacity utilization + waitlist conversion	Premium tier + standard tier — captures both, premium has 6-mo waitlist creating prestige

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named platform must be real (Brightwheel, Procare, HiMama, Kangarootime, Smartcare)
- Every named curriculum / cert must be real (AMS, AMI, NAMTA, NAEYC, Waldorf)
- Every named subsidy must be real (state-specific: WI Shares, CA CDE, etc.)
- Every named corporate backup partner must be real (Bright Horizons, Care.com, KinderCare)

6. Cost scale (L5 — per-sector currency bands)

Child care center spend bands are small-business + facility scale:

1 — Under \$500	Subsidy enrollment, employer outreach, app trial
2 — \$500 to \$5,000	Software setup, signage, marketing, NAEYC application
3 — \$5,000 to \$50,000	Curriculum cert + training, outdoor refresh, programming
4 — \$50,000 to \$500,000	Major facility upgrade, classroom expansion
5 — Over \$500,000	New facility, second location, full buildout

7. Persona schema (L6)

Schema: B2C_consumer (parent + family decision)

Required persona fields:

- Parent name + child age(s) + family composition
- Working schedule (full-time, part-time, both parents work)
- Curriculum preference + values
- Budget tier + employer benefit availability
- Acquisition channel (search, friend, employer, drive-by)
- LTV (years × monthly tuition × siblings)

8. Cycle model (L7 — replaces calendar seasons)

Child care runs on academic + new-year + summer cycles:

Enrollment season (Feb-Apr)	Fall enrollment decisions	Maximum tour + open house cadence	
Late summer (Aug)	Last-minute fall enrollment surge	Final tours + emergency enrollment	
Fall (Sep-Nov)	Settled enrollment + waitlist building	Family events + community building	
Winter (Dec-Feb)	Re-enrollment + budget planning	Re-enrollment commitment + tuition review	
Spring (Mar-May)	Summer camp registration + transitions	Summer camp enrollment + graduation events	

Summer (Jun-Aug)	Camp programs + smaller enrollment	Camp programs + facility maintenance
Annual: licensing + renewal accreditation	State + accreditation calendar	Compliance walkthrough + renewal filings

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Teacher turnover impact on kids — universal — frame as retention investment
- Communication gaps with parents — universal — frame as app + daily report
- Sick-day exclusion stress — universal — frame as backup care + clear policy
- Pricing surprise / annual increases — universal — frame as upfront annual schedule
- Drop-off / pick-up traffic chaos — universal — frame as curbside + drive-thru protocol
- Outdoor space limitations — universal — frame as off-site nature program
- Accreditation pressure — universal — frame as multi-year roadmap

10. Report sections (L9 — section list and order)

- 1 **Center snapshot** — Capacity, enrollment, age groups, named comp set
- 2 **Top 5 growth + retention moves** — Drawn from 14-category playbook
- 3 **Curriculum + accreditation roadmap** — NAEYC + Montessori-style positioning
- 4 **Pricing + payer-mix strategy** — Subsidy + employer + private mix
- 5 **Family personas (4 + lost family)** — Working family / curriculum-driven / subsidy / corporate
- 6 **Enrollment + waitlist plan** — Annual enrollment cycle + tour discipline
- 7 **Annual cycle plan** — Academic + summer + licensing calendar
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Enrollment count + capacity utilization %
- Tuition rate + payer mix
- Parent retention 12-mo (target > 80%)
- Teacher tenure (avg, target > 2 yr)
- NAEYC accreditation status
- Waitlist depth by age group

Out-of-scope NAICS codes inside this sector

These codes belong to this sector but BizRadar refuses to analyze them — either because they are not real businesses, are regulated monopolies BizRadar cannot help, or do not have customer-facing operations to optimize.

62211x	General hospitals — too large + B2G regulatory; BizRadar value-add minimal
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622210	Psychiatric + substance abuse hospitals — regulated specialty
623110-623990	Nursing + residential care — covered by adjacent profile (queued for separate sub-cluster)
624190-624230	Social assistance / community / emergency / housing — mostly nonprofit, not BizRadar fit
624310	Vocational rehabilitation — nonprofit / government-funded

Engineering note for this sector

Sector 62 is the highest-LTV B2C-appointment expansion for BizRadar. Build healthcare.medical_practice first (Phase 1, biggest market), then dental (Phase 1, second largest), then mental health (Phase 1, fastest-growing), then child care (Phase 2). Add nursing / home health / residential care as a Phase 5 sub-cluster — they need different B2B-influenced logic.

CHAPTER 71 — ARTS, ENTERTAINMENT, RECREATION

Fitness studios, attractions, museums, performing arts, casinos, country clubs, golf, ski. Mostly fits v3 architecture but splits into 3 distinct profiles.

Covers 25 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.

The v3 problem in this sector

A yoga studio fits v3 perfectly. A museum needs membership / endowment / education-program logic v3 lacks. A performing-arts venue runs on subscription seasons + grants. Three profiles required.

Sub-clusters in this sector

This sector splits into 2 sub-clusters because the business models inside diverge enough that one profile cannot serve them all. Each sub-cluster gets its own complete spec below.

recreation.fitness_studio	Fitness centers, yoga / pilates / barre / climbing studios, dance, martial arts	B2C_appointment	713940, 611620 (sports instruction), 812199 (other personal care related to wellness)
recreation.attraction	Museums, zoos, amusement parks, country clubs, golf courses, ski areas, marinas	B2C_walk_in	712110, 712120, 712130, 712190, 713110, 713920, 713910, 713930

71.fitness_studio — Fitness centers, yoga / pilates / barre / climbing studios, dance, martial arts

Business model: B2C_appointment | **Profile ID:** recreation.fitness_studio

Covers NAICS: 713940, 611620 (sports instruction), 812199 (other personal care related to wellness)

Modern formats (Part 2): Climbing Gym, CrossFit Box, Pilates / Reformer Studio, Dance Studio, Martial Arts Studio, Yoga Studio

1. Data sources (replaces v3 hospitality stack)

Google Places + Reviews	Reviews, photos, hours	\$0.02	Yelp
Yelp Fusion API	Reviews + categories	\$0.01	Manual

ClassPass + MindBody (consumer side)	Class booking demand patterns	Free observe	Direct call
MindBody / Glofox / Vagaro benchmarks	Industry retention + ATV benchmarks	Subscription public	Manual
IHRSA reports	Fitness industry benchmarks	Subscription	Public summaries
Foursquare / SafeGraph	Foot traffic + dwell	Subscription	Door count
Strava / Apple Fitness (community)	Local fitness community signals	Free	Manual scrape
Local school + corporate wellness	Named potential corporate / school clients	Free	Direct call

2. Competitor radius

3 km urban, 8 km suburban, 20 km rural. Boutique studios compete by modality (yoga vs barre vs HIIT) within 5 km. Big-box gyms (Planet Fitness, LA Fitness) considered separate competitor class.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Instructor quality + variety
- Class schedule fit
- Equipment / facility quality
- Cleanliness
- Locker room + showers
- Booking + cancellation flexibility
- Community / vibe
- Pricing / membership transparency

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Member retention + community programming	Industry benchmark retention rate (target > 75% annual)	Member challenges + monthly social — retention from 64% to 78% — adds \$48k LTV on 200-member base
Class pass / drop-in program	ClassPass partnership economics + ASP	ClassPass partnership — \$48 commission per drop-in — fills underused 6am + 2pm slots
Corporate wellness contract	Named local employers + HR wellness budgets	Named 200-employee software co. — \$14/employee/mo all-access — \$33.6k/yr predictable revenue

Workshop / specialty series	Modality demand + capacity + ASP	6-week pre-natal yoga series — \$180/student × 12 students × 4 cycles/yr = \$34.5k zero-equipment revenue
Teacher training / certification program	Yoga Alliance / NASM certifications + program ASP	200-hr YTT — \$2,800/student × 8 students/yr = \$22.4k high-margin + builds future teachers
Retail merch + nutrition add-on	POS-integrated retail + supplier relationships	Branded apparel + protein bars — 14% of revenue at 50% margin = \$3,500/mo incremental
Personal training / 1-on-1 add-on	PT pricing benchmark + capacity	\$95/session × 60 sessions/wk × 60% margin to studio = \$148k/yr studio share
Social media + influencer-instructor program	IG + TikTok creator economics in fitness	Named lead instructor builds personal brand — drives 18% of new memberships from social referral
Free intro / new-student funnel	Conversion benchmark from intro to paying member	"7-day intro for \$14" — 22% conversion to membership — 4× cheaper than paid ads
Loyalty / referral program	Referral incentive structures + benchmark	"Refer-a-friend, get 1 month free" — 14% of new members come from referrals
Live / on-demand digital revenue	Online platform economics (Glofox, Vimeo OTT)	\$24/mo digital-only tier — 80 subscribers = \$23k/yr from previously-churned members
School / sports-team partnership	Named local schools + sports programs	High-school cross-country team yoga — \$200/session × 18 sessions = \$3,600 + brand exposure
Charity / community event hosting	Named local nonprofits + community demand	Quarterly charity yoga class — \$20 entry → food bank — drives 80 new attendees per event
Class schedule optimization	Demand by daypart vs current schedule	6:15am + 12:00pm + 5:30pm classes filled to 90% capacity, weekend gap analysis adds 4 classes/wk

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named platform must be real fitness software (MindBody, Glofox, ClubReady, Vagaro, Pike13)
- Every cert must be a real org (Yoga Alliance, NASM, ACE, ACSM, NSCA)
- Every named local partner must be a verifiable employer / school
- Every benchmark must come from IHRSA / MindBody / industry source

6. Cost scale (L5 — per-sector currency bands)

Fitness studio spend bands are small-business hospitality scale:

1 — Under \$200	Schedule change, partner outreach, social posting
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2 — \$200 to \$2,000	Workshop launch, retail buy-in, photo shoot
3 — \$2,000 to \$20,000	Software upgrade, equipment refresh, intro campaign
4 — \$20,000 to \$100,000	Studio renovation, new equipment line, second location
5 — Over \$100,000	New location, full buildout, expansion

7. Persona schema (L6)

Schema: B2C_consumer (with retention lifecycle fields)

Required persona fields:

- Name + age + fitness goal
- Frequency target (3×/wk, 5×/wk)
- Modality preference + cross-modality willingness
- Acquisition channel (intro promo, IG, friend, walk-by)
- Retention risk + lifecycle stage
- LTV (months retained × monthly fee + add-ons)

8. Cycle model (L7 — replaces calendar seasons)

Fitness studios run on resolution + summer + back-to-school cycles:

New Year (Jan)	Resolution surge — 30-40% of new memberships	Capacity planning, intro promo, retention focus on cohort
Spring (Mar-May)	Summer-body push	Specialty program, before/after photo program
Summer dead zone (Jun-Aug)	Travel + outdoor competition	Outdoor classes, vacation hold policies, social events
Back-to-school (Sep)	Schedule reset surge	Routine reset campaign, schedule changes
Holiday slowdown (Nov-Dec)	Engagement drop	Community events, gift-card sales, locked-in renewals
Weekly: peak hours	6am, noon, 5-7pm, weekend AM	Capacity + instructor scheduling discipline
Annual: instructor reviews + pay	Retention of best teachers	Tenure-based pay + revenue share with senior teachers

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Booking system frustration — universal — frame as platform UX + customer service
- Class no-show / waitlist — universal — frame as deposit + waitlist policy

- Instructor turnover — universal — frame as career path + revenue share
- Locker / shower availability — universal — frame as facility scheduling
- Membership cancellation surprise — universal — frame as transparent cancel policy
- Big-box price competition — universal — frame as community + instructor moat

10. Report sections (L9 — section list and order)

- 1 **Studio snapshot** — Modality, members, ASP, retention, named comp set
- 2 **Top 5 growth + retention moves** — Drawn from 14-category playbook
- 3 **Member lifecycle plan** — Acquisition → onboarding → retention → win-back
- 4 **Programming + schedule optimization** — Demand-by-daypart + new-class proposals
- 5 **Customer personas (4 + lost member)** — New / regular / VIP / churned
- 6 **Annual cycle plan** — 7-window calendar with action items
- 7 **Comp set + pricing comparison** — Named competitors with offer structure
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Active member count + retention rate (12-mo)
- Average revenue per member (ARPM)
- Class capacity utilization %
- Intro-to-member conversion rate (target > 20%)
- Net promoter score (target > 50)
- Add-on revenue % (PT, retail, workshops)

71.attraction — Museums, zoos, amusement parks, country clubs, golf courses, ski areas, marinas

Business model: B2C_walk_in | **Profile ID:** recreation.attraction

Covers NAICS: 712110, 712120, 712130, 712190, 713110, 713920, 713910, 713930

1. Data sources (replaces v3 hospitality stack)

Google Places + TripAdvisor	Reviews + ranking + photos	\$0.02	Yelp	
State / regional tourism office	Visitor stats + named circuit partnerships	Free	Direct call	
IAAPA / AZA / AAM benchmarks	Industry attendance + ASP benchmarks	Member-only	Public summaries	
SafeGraph / Foursquare	Visit pattern + dwell time	Subscription	Door count	
Eventbrite / Facebook Events	Local + regional event calendar	Free tier	Manual	
Census household + family data	Target demographic by ZIP	Free	Direct	
NOAA + weather	Seasonal demand forecasting	Free	Internal	
Local school district	Named school field-trip programs	Free	Direct call	

2. Competitor radius

60-90 mi for destination attractions (museums, amusement, zoos). 20 mi for daily-recreation (mini-golf, bowling). 100+ mi for unique niche (only zoo / only ski). Country clubs / golf compete within 25 mi.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Value for money
- Wait times for rides / experiences
- Cleanliness / restrooms
- Food + concessions
- Family / kid friendliness
- Educational content (museums)
- Animal welfare (zoos)
- Course condition (golf)
- Snow conditions / lift wait (ski)
- Staff friendliness

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Membership / season pass program	Industry membership economics (zoos, museums, ski)	\$95 family annual pass — 1,400 sold = \$133k upfront cash + 4× normal visit frequency
School field trip program expansion	Named school districts + curriculum mapping	4 named districts × 30 buses × 50 kids × \$9/kid = \$54k seasonal revenue
Birthday party / event packages	Local family party benchmarks	\$320 birthday package × 80 parties/yr = \$25.6k easy add-on revenue
Wedding / private event venue	County zoning + venue benchmark	Lakeside event rental — \$4,200 minimum — 14 events/yr = \$59k incremental
Educational + adult programming	Adult-learner demand + university partnerships	Wine + paint at art museum — \$48/person × 24 people × 12 events = \$13.8k
Member-only + early-access tier	Loyalty + tiered membership economics	"Premier" \$250 tier — 320 members = \$80k upfront + creates VIP perks
Nonprofit grant + endowment growth	Named foundations + state arts council grants	NEA Challenge America \$25k + Wisconsin Arts Board \$45k + capital campaign launch
Restaurant / food upgrade	Per-cap food spend benchmark	Refresh food vendor partner — per-cap food spend from \$4.20 to \$7.80 — adds \$720k/yr at 100k visits
Photo + memorabilia revenue	Per-cap retail spend benchmark	On-ride photo + queue gift shop — \$4.80 per-cap incremental
Off-season programming	Named off-season events + capacity	Holiday-light walkthrough at zoo Nov-Jan — \$18 entry — captures 60k visitors at \$1.1M zero-animal-cost
Tourist circuit partnership	Named regional tourism circuit	WI Cheese Trail / Driftless circuit — listed on state map — adds 18% tourist visits
Sponsorship / corporate partnership	Named local + regional brand sponsors	Named regional bank — \$40k naming sponsor for new exhibit — recurring annual
Capital campaign / new exhibit	Major donor + grant landscape + match programs	\$2M new exhibit campaign — 60% donor + 40% grant match — drives 18% attendance lift
Reservation / timed-entry shift	Industry move post-COVID + capacity management	Timed-entry tickets — improves dwell time + per-cap by 14% via crowd dispersion

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named foundation / grant must be real (NEA, NEH, IMLS, state arts council)
- Every named circuit / partner must be currently operating
- Every benchmark must come from AAM / AZA / IAAPA / NRPA
- Every named sponsor / corporate partner must be a verifiable regional company

6. Cost scale (L5 — per-sector currency bands)

Attractions / venues operate at small-org institutional bands:

1 — Under \$500	Programming change, membership tier, partnership outreach
2 — \$500 to \$5,000	Marketing campaign, photo shoot, school district pitch
3 — \$5,000 to \$50,000	New exhibit (small), event venue setup, technology upgrade
4 — \$50,000 to \$500,000	Major exhibit, capital improvement, restaurant rebuild
5 — Over \$500,000	Capital campaign, new wing, major animal habitat

7. Persona schema (L6)

Schema: B2C_consumer (family + repeat-visitor lens)

Required persona fields:

- Name + family composition + ages
- Distance willing to drive
- Visit-trigger (birthday, school break, weather, named event)
- Member vs single-visit propensity
- Photo / share behavior
- LTV (visits/yr × ticket × food + retail attach)

8. Cycle model (L7 — replaces calendar seasons)

Attractions run on weather + school + holiday cycles:

Summer (Jun-Aug)	Peak — 40-60% of annual attendance	Maximum operations, summer programming, hire seasonal staff
Spring (Mar-May)	Spring break + Mother's Day + school trips	School field trip surge, spring programming
Fall (Sep-Oct)	Foliage + harvest events	Halloween / fall programming, seasonal exhibits
Holiday season (Nov-Dec)	Light shows + holiday markets	Major holiday programming + gift-membership push
Winter (Jan-Feb)	Slowest — except ski / indoor	Member retention + gala / capital campaign timing
School breaks	Spring break / winter break / summer break	Targeted marketing 4 weeks ahead
Weekly: weekend surge + weekday slow	Capacity asymmetry	Weekday pricing + weekend capacity management

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Long wait / line complaints — universal — frame as virtual queue + capacity
- Restroom availability — universal — frame as cleaning protocol + count
- Concession pricing complaints — universal — frame as bundled or quality positioning
- Parking / access — universal — frame as shuttle + paid lot management
- Animal welfare critiques (zoos) — universal — frame as transparency + accreditation (AZA)
- Course condition / snow / wave (golf, ski, marina) — frame as transparency + comm protocol
- Member-perceived value erosion — universal — frame as exclusive benefits

10. Report sections (L9 — section list and order)

- 1 **Venue snapshot** — Annual visits, ASP, members, named comp set
- 2 **Top 5 strategic + revenue moves** — Drawn from 14-category playbook
- 3 **Membership + retention plan** — Tiered offer + benefit map
- 4 **School + group sales pipeline** — Named district + corporate prospects
- 5 **Sponsorship + grant pipeline** — Named funders + capital priorities
- 6 **Customer personas (4 + lost visitor)** — Family / member / tourist / school
- 7 **Annual cycle plan** — 7-window calendar
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Annual visits + per-cap spend
- Member count + member visit %
- School field trip bookings (count + revenue)
- Event / wedding revenue % of total
- Per-cap food + retail attach
- Earned revenue % vs grants / endowment (target > 60%)

Engineering note for this sector

Sector 71 is mostly a v3-format port for fitness studios, but attractions / museums / venues need new logic for membership tiers, school programs, and grant pipelines. Build recreation.fitness_studio first (Phase 1, easy port), then recreation.attraction (Phase 1/2, new logic).

CHAPTER 72 — ACCOMMODATION & FOOD SERVICES

Hotels, restaurants, bars, food trucks, catering. The v3 sweet spot — the entire BizRadar pipeline was built around this sector.

Covers 15 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.

The v3 problem in this sector

v3 already works for this sector. The v4 work here is small: lift the existing hospitality logic out of server.js and re-shape it as a sector_profile so the rest of the pipeline can swap it for other sectors. No new analysis is needed — only refactoring.

Sub-clusters in this sector

This sector splits into 3 sub-clusters because the business models inside diverge enough that one profile cannot serve them all. Each sub-cluster gets its own complete spec below.

Sub-cluster	Description	Business Model	NAICS Codes
hospitality.lodging	Hotels, motels, B&Bs;, RV parks, vacation rentals	B2C_walk_in	721110, 721120, 721191, 721199, 721211, 721214, 721310
hospitality.restaurant	Restaurants (full-service, limited-service, ghost kitchen)	B2C_walk_in	722511, 722513, 722514, 722515
hospitality.bar_brewery	Bars, breweries, taprooms, distilleries with on-site service	B2C_walk_in	722410, 722515, 312120 (brewery taproom), 312140 (distillery taproom)

72.lodging — Hotels, motels, B&Bs;, RV parks, vacation rentals

Business model: B2C_walk_in | **Profile ID:** hospitality.lodging

Covers NAICS: 721110, 721120, 721191, 721199, 721211, 721214, 721310

Modern formats (Part 2): Boutique Hotel, Glamping Site, Hostel, Vacation Rental Operator (multi-unit Airbnb)

1. Data sources (replaces v3 hospitality stack)

Data Source	Description	Cost	Notes
Google Places + Reviews	Reviews, photos, ratings, hours	\$0.02	Yelp scrape
TripAdvisor	Trip-specific reviews + ranking	\$0.01	Booking.com

Booking.com / Expedia	Pricing, availability, OTA reviews	Scrape	STR data
STR / CoStar lodging data	ADR / RevPAR / occupancy benchmarks	Subscription	AHLA reports
Yelp Fusion API	Local categorization + reviews	\$0.01	Manual scrape
Foursquare	Foot traffic patterns + tip data	\$0.005	SafeGraph
VisitX / state tourism office	Named regional events + tourist stats	Free	Direct call
NOAA + local event calendars	Weather + festival demand drivers	Free	Eventbrite

2. Competitor radius

2.5 km for urban, 8 km for suburban, 25 km for rural / destination. Independent hotels compete with branded chains in the same radius — do not filter by flag.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Cleanliness
- Bed comfort
- WiFi quality
- Front desk service
- Location convenience
- Breakfast / amenities
- Noise / quietness
- Bathroom + shower quality
- Value for money
- Parking ease

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **18** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Sensory experience layer	Named local scent / soap / coffee makers within 60 mi	Replace generic shampoo with locally-made Tropi Bath Co. amenities — \$1.40/room incremental cost, lifts amenity rating
Photo-moment installation	Local instagram trends + visiting demographic	Lobby mural by named local artist — \$2,800 — generates ~80 IG tags/month
Hyper-local breakfast partnership	Named local bakeries / dairies / coffee roasters within 60 mi	Carr Valley Cheese plate at breakfast bar — \$2.20/guest premium cost, sells \$14 cheese boards in lobby

Local-experience concierge layer	Named experiences / outfitters / events within 30 mi	Curated list of 8 named experiences (Governor Dodge State Park hike map, House on the Rock, Carr Valley factory tour) printed in-room
Ancillary revenue: pet fee, late checkout, parking	Comp set ancillary pricing benchmarks	Late checkout to 2pm at \$35 — captures \$4,800/yr at 30% of guests on shoulder days
Direct booking premium	OTA commission % vs direct conversion lift levers	Free upgrade for direct bookers — saves 18% OTA commission + drives 12% repeat rate
Group / wedding / event business	Named local venues + wedding planners + corporate clients	Marriage license room block at Iowa County Courthouse — 24 weddings/yr × 12 rooms × \$129 = \$37k
Loyalty + repeat-stay layer	Email list size + open rate + offer mechanics	"Stay 3 nights, 4th free" emailed to past guests — captures shoulder-season rebooking, 8% conversion
Workforce / extended-stay rate	Named regional employers with travelers	Foxconn build crew preferred rate — 2,200 room-nights/yr at \$89 vs walk-in \$129 = \$264k revenue
Service-quality scoring program	Mystery-shop services + monthly QA	Quarterly LRA Mystery Shop — \$480/visit — ID specific service gaps before guests post negative review
Air B&B; alternative positioning	STR ban / regulation status + price gap	Run a "no cleaning fee" promo vs Airbnb — captures 22-night-stay segment fleeing inflated STR fees
Sustainability / EV charging story	NEVI fund + state rebate + EV demand	Install 4 ChargePoint Level 2 stations — \$14k after Wisconsin grant — adds 6% EV-driver bookings
Off-season packaging	Named regional events + low-occupancy windows	Driftless Winter Photo Tour package — Jan-Feb — \$189 includes room + photographer + breakfast
Brand affiliation analysis	Soft-brand vs flag economics	Switch from Best Western to Ascend Collection (Choice soft-brand) — keeps independence, lifts ADR \$14
Guest-data + email list build	CRM stack + opt-in mechanics	Pre-arrival upsell email — 3,400 opens/yr × 4% conversion × \$42 avg = \$5,700/yr
Photo / video refresh on listings	OTA photo conversion benchmarks	Hire named local photographer — \$1,800 shoot — TripAdvisor ranking lifts 8 spots, +6% bookings
Hyper-local content marketing	Local blog / news / Reddit presence	Monthly "what's open in Dodgeville this week" newsletter — builds local authority + repeat traffic

Workforce retention + culture	Local wage benchmark + retention cost	Pay \$1.50/hr above local benchmark — turnover drops from 110% to 35% — saves \$48k/yr in re-hiring
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5. Specificity rules (L4 — what 'real named entity' means here)

- Every named producer / partner must exist in Google Places within 60 mi
- Every named event must have a verifiable date + URL
- Every benchmark (ADR, RevPAR, occupancy) must reference STR or AHLA published data for this market tier
- Every named brand affiliation option must be a real soft-brand / flag program

6. Cost scale (L5 — per-sector currency bands)

Lodging operators evaluate spend in mid-range hospitality bands:

1 — Free or under \$200	Process change, partner outreach, listing refresh
2 — \$200 to \$2,000	Photo shoot, signage, amenity test, mystery shop
3 — \$2,000 to \$20,000	Lobby refresh, EV station, brand affiliation cost
4 — \$20,000 to \$100,000	Room renovation, PMS upgrade, breakfast remodel
5 — Over \$100,000	Full renovation, addition, branded conversion

7. Persona schema (L6)

Schema: B2C_consumer (the existing v3 persona schema)

Required persona fields:

- Name + age + life stage
- Travel context (business, leisure, family, milestone)
- Distance willing to drive + transit mode
- Spend trigger (what makes them book this property over alternatives)
- 5-star trigger (what earns the great review)
- Disappointment trigger (what makes them post a 1-star)
- Lifetime value (stays/yr × ADR × nights)
- Acquisition channel (named OTA, search query, referral source)

Example persona: David, 52, business traveler — quarterly Foxconn site visits

Drives from Milwaukee, 90 min. Spend trigger: USB outlets at desk, predictable WiFi, late checkout for afternoon flight. 5-star trigger: hot breakfast before 6am with quality coffee. 1-star trigger: noisy hallway from late check-ins. LTV: 4 stays/yr × 2 nights × \$129 = \$1,032/yr × 4 yrs = \$4,128.

8. Cycle model (L7 — replaces calendar seasons)

Lodging follows leisure + corporate seasonal patterns:

Pre-summer (Apr-May)	Spring break + Mother's Day demand	Activate family packages, refresh outdoor amenities
Summer peak (Jun-Aug)	Tourist + family travel	Maximum ADR, capacity management, OTA optimization
Shoulder (Sep-Oct)	Fall foliage + business travel resumption	Group / corporate sales push
Holiday season (Nov-Dec)	Family reunions + holiday parties	Holiday packages, group dining contracts
Deep off (Jan-Feb)	Lowest occupancy	Winter packages, locals stay-cation, maintenance projects
Spring (Mar)	Wedding season ramp	Wedding party blocks, prom / event sales
Annual: brand standards audit (if franchised)	Flag compliance review	Pre-audit walk-throughs, capex prioritization

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- WiFi reliability — universal — frame as infrastructure quick-win
- Cleanliness perception — universal — frame as housekeeping audit
- Front-desk attitude variance — universal — frame as hiring + culture
- Bed comfort age — universal — frame as 7-yr replacement cycle
- Breakfast quality vs OTA expectations — universal — frame as named-local upgrade
- Parking confusion — universal — frame as wayfinding + email pre-arrival
- Noise from highway / next room — universal — frame as soundproofing + room blocks

10. Report sections (L9 — section list and order)

- 1 **Property snapshot** — Rooms, ADR, occupancy, RevPAR vs comp set, named events nearby
- 2 **Top 5 unique experiences to add** — Drawn from 18-category playbook with named local partners
- 3 **Quick wins (under \$2k)** — Process changes + partner outreach with named entities
- 4 **Hyper-local partnership map** — Named producers / experiences / brands within 60 mi
- 5 **Customer personas (4 personas + lost guest)** — Business / leisure / family / milestone
- 6 **Cycle plan** — 7-window calendar with action items
- 7 **Comp-set comparison** — Named competitors with their gaps and strengths
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- ADR vs comp set (\$)
- Occupancy % vs comp set
- RevPAR vs comp set

- TripAdvisor / Google rating vs comp set
- Direct booking % (target > 35%)
- Repeat-guest rate (target > 20%)

72.restaurant — Restaurants (full-service, limited-service, ghost kitchen)

Business model: B2C_walk_in | **Profile ID:** hospitality.restaurant

Covers NAICS: 722511, 722513, 722514, 722515

Modern formats (Part 2): Ghost Kitchen, Pop-Up Restaurant, Chef's Counter / Omakase, Fast-Casual

1. Data sources (replaces v3 hospitality stack)

Google Places + Reviews	Reviews, photos, hours, menu	\$0.02	Yelp	
Yelp Fusion API	Reviews + photo OCR + categories	\$0.01	Manual	
TripAdvisor	Visitor reviews + ranking	\$0.01	OpenTable	
OpenTable / Resy / Tock	Booking demand patterns + dwell time	Subscription scrape	/ Reservation logs	
DoorDash / UberEats / GrubHub	Delivery menu + pricing + ratings	Scrape	Direct API	
Restaurant Business / Datassential	Industry trend + menu intelligence	Subscription	NRA reports	
SafeGraph / Foursquare	Foot traffic + dwell time	Subscription	Manual count	
Local food blogger + social	Local press + influencer mention	Free	Manual search	

2. Competitor radius

1 km urban, 4 km suburban, 12 km rural. For destination dining, expand to 25+ km. For ghost kitchens / delivery-only, the radius is delivery-area-defined.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Food taste / quality
- Service speed
- Server attentiveness
- Portion size vs price
- Ambiance + decor
- Cleanliness
- Menu variety
- Drink program
- Noise level
- Reservation / wait management

- Take-out / delivery quality

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **18** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Hyper-local sourcing storytelling	Named farms / dairies / butchers within 60 mi	Menu naming: "Steaman Farm dry-aged ribeye" — adds \$4 menu price acceptance + Instagram-shareable	
Tasting / chef's counter experience	Local fine-dining ASP + dinner-experience demand	\$95 7-course chef's counter — Wed/Thu nights — 12 seats × 2 turns = \$2,280/night	
Brunch / weekend pivot	Weekend demand vs weekday + brunch ASP	Add Sat-Sun 10-2 brunch — incremental \$7,400/wk at 60% margin = \$4,440 contribution	
Private dining / buyout offering	Named local corporate clients + event planners	14-seat private room — \$1,200 minimum F&B; — 18 events/yr = \$21,600 incremental	
Cooking class + retail jar program	Local food enthusiast demand + capacity	\$95/person × 12 people × 8 classes/yr = \$9,120 + \$14 jars sold to attendees	
Beverage program upgrade	Local distillery / brewery / coffee roaster partners	Featured cocktail with named local distillery — \$3 BOGO promotion lifts cocktail mix 18%	
Loyalty + birthday club	POS-integrated CRM (Toast, Square, Shift4)	Birthday email with \$20 gift — 2,200 emails × 18% redeem × \$58 avg ticket = \$23k incremental	
Catering + drop-off offering	Local corporate / school catering market	School lunch contract — 2 named districts — 600 meals/wk × \$9 = \$5,400/wk	
Influencer / press dinner program	Named local food bloggers + journalists	6 comp dinners to "Madison Eats" + "Madison Magazine" food editor — drives 3-month traffic lift	
Menu engineering audit	Item profitability + popularity matrix	Boston Matrix exercise — promote 4 "stars," redesign 3 "puzzles," 86 the 2 "dogs"	
Online ordering on own domain	Toast / ChowNow / Square Online setup	Direct ordering on own site — saves 22% delivery commission on 40% of online orders	
Outdoor seating expansion	County permit + season length + extra-cover analysis	14-seat patio — Apr-Oct — 24 covers/day × \$48 ASP × 35% margin = \$14k incremental contribution	
Wine list re-build with local sommelier	Local wine reps + storage capacity	Curated 24-bottle list with named distributor — wine attach lifts from 22% to 38% of tickets	
Server training program	Named training programs (Rouxbe, BarSmarts)	4-week training program — guest check increases avg \$2.40, 240 covers/wk = \$576/wk × 52 = \$30k/yr	

Off-premise dinner kit / heat-and-serve	Local family-meal demand + packaging cost	Sunday family-meal dinner kit — \$48 serves 4 — 60 kits/Sun × 52 wk = \$150k incremental
Reservation strategy + walk-in mix	Resy / OpenTable seat-yield data	70% reservation / 30% walk-in target — increases average occupancy from 62% to 81%
Photo / menu refresh on Google + OTA	Photo conversion benchmark on Google	\$1,200 named photographer — Google profile views lift 47% — calls-from-search up 28%
Festival / event participation	Named regional food festivals + cost / ROI	Madison Restaurant Week — \$0 entry — captures 2,800 first-time guests at \$35 prix fixe

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named producer / partner must exist in Google Places or state ag directory
- Every benchmark must reference NRA / Datassential / state hospitality association
- Every named blogger / press must be a real outlet currently publishing
- Every catering / event partner must be a verifiable local org

6. Cost scale (L5 — per-sector currency bands)

Restaurant operators evaluate spend in restaurant operating bands:

1 — Free or under \$200	Menu copy, photo refresh, partnership outreach
2 — \$200 to \$2,000	Server training, signage, photo shoot, small equipment
3 — \$2,000 to \$20,000	Outdoor seating, beverage program rebuild, POS upgrade
4 — \$20,000 to \$100,000	Kitchen equipment, dining room refresh, wine cellar build
5 — Over \$100,000	Full renovation, second location, kitchen rebuild

7. Persona schema (L6)

Schema: B2C_consumer

Required persona fields:

- Name + age + occasion type
- Spend trigger (what unlocks higher ticket)
- Frequency / repeat probability
- Photo behavior (does this person share)
- Group composition (date / family / business / solo)
- Acquisition channel (Google search, IG, friend referral, OTA)

- LTV (visits/yr × ASP)

Example persona: Lauren, 31, dual-income foodie — Madison

Drives 50 min for "an experience". Spend trigger: tasting menu + named-local sourcing story. Photo behavior: posts every dish to IG @lauren.eats (8k followers). Tells 3-4 friends per visit. LTV: 6 visits/yr × \$124 ticket = \$744/yr × 5 yrs = \$3,720.

8. Cycle model (L7 — replaces calendar seasons)

Restaurants run on weekly + seasonal + holiday cycles:

Sun-Wed lulls	Weekday slow nights	Industry night, locals night, prix fixe specials
Thu-Sat peak	Highest-volume window	Reservation discipline, runner support, expeditor doubling
Brunch (Sat-Sun)	Weekend daytime demand	Distinct menu, batch prep, mimosa program
Holiday peaks (Mother's Day, NYE, V-Day)	Booked weeks ahead	Prix fixe, prepay deposits, seating maximization
Restaurant Week / festival windows	City-led event participation	Curated \$35/\$45 prix fixe, captures new guests
Patio season (Apr-Oct in north)	Outdoor capacity unlock	Hire seasonal patio staff, maintenance pre-season
Off-season menu refresh (Jan, Aug)	Slow-down + new ideas	Seasonal menu + new server training

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Slow service on busy nights — universal — frame as expeditor / runner / pre-bus
- Inconsistent food quality — universal — frame as line training + recipe cards
- Long wait without reservation — frame as Resy / waitlist app
- Loud noise level — universal — frame as acoustic panels + music level
- Menu price perception — universal — frame as portion / story / value bundling
- Server attitude variance — universal — frame as hiring + tipping pool design
- Reservation no-shows — universal — frame as deposit / credit-card hold

10. Report sections (L9 — section list and order)

- 1 **Restaurant snapshot** — Cuisine, average ticket, covers/wk, top reviews, named comp set
- 2 **Top 5 unique menu / experience plays** — Drawn from 18-category playbook
- 3 **Quick wins (under \$2k)** — Process changes + named partner outreach
- 4 **Hyper-local sourcing map** — Named producers / partners within 60 mi
- 5 **Customer personas (4 personas + lost guest)** — Date / family / business / solo
- 6 **Weekly cycle plan** — 7-window calendar with daypart actions

7 **Menu engineering scoreboard** — Boston Matrix on top 20 items

8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Avg ticket per cover (\$)
- Covers per week + occupancy %
- Beverage attach % (target > 30%)
- Reservation no-show rate (target < 4%)
- Google + Yelp avg rating (target > 4.5)
- Off-premise % of revenue (delivery + catering)

72.bar_brewery — Bars, breweries, taprooms, distilleries with on-site service

Business model: B2C_walk_in | **Profile ID:** hospitality.bar_brewery

Covers NAICS: 722410, 722515, 312120 (brewery taproom), 312140 (distillery taproom)

Modern formats (Part 2): Microbrewery + Taproom, Cidery, Meadery, Cocktail Bar, Wine Bar

1. Data sources (replaces v3 hospitality stack)

Google Places + Reviews	Reviews, photos, hours	\$0.02	Yelp	
Untappd	Beer-specific reviews + check-ins	API	Manual scrape	
Distiller / Vivino	Spirits / wine ratings	Free tier	Manual	
BeerAdvocate / RateBeer	Brewery community sentiment	Free	Manual	
Brewers Association directory	Named regional + craft peers	Member-only	State guild	
State alcohol beverage commission	License status + violations	Free	Direct office	
Eventbrite / Facebook Events	Local event calendar competition	Free tier	Manual	
Foursquare	Foot traffic + dwell time	Subscription	POS data	

2. Competitor radius

1 km urban, 4 km suburban, 15 km rural. Destination breweries (e.g. New Glarus) draw 60+ km. Distilleries / wineries with tasting rooms similar to wineries — 25+ km destination radius.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Beer / drink quality
- Variety / rotating taps
- Bartender knowledge + speed
- Atmosphere / vibe
- Food (if served)
- Live music / events
- Outdoor / patio
- Dog / kid friendliness
- Wait at bar
- Pricing transparency

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Tap takeover / featured guest tap	Named regional breweries + distribution agreements	Monthly Lakefront Brewery tap takeover — drives 35% Saturday-night attendance lift
Member / mug club program	Successful peer programs + member benefit menu	\$60 annual mug club — 320 members = \$19k upfront + 4× normal frequency
Live music / trivia / event calendar	Local musicians + trivia leagues + capacity	4 nights/wk programming — Tue trivia, Thu open mic, Fri/Sat live — fills slow weekdays
Food truck rotation partnership	Named local food trucks + permit terms	5-truck rotation — Wed-Sun — taproom keeps food sales hassle-free, drivers \$80 fuel + tips
Crowler / growler off-premise sales	Self-distribution license + can / fill cost	On-site canning — \$14/4-pack — 160 packs/wk × 52 = \$116k off-premise revenue at 65% margin
Distribution expansion	Named distributors + state self-distribution rules	Add 8 named accounts via Beer Capitol Distributing — \$48k/mo wholesale at \$128/keg
Brewery tour + tasting program	Tour pricing benchmark + capacity	\$22 tour + 6-pour flight — Sat 11am, 1pm, 3pm — 60 guests × 3 × \$22 = \$3,960/Sat
Private events / brewery rental	Named local corporate / wedding clients	Brewery wedding rental — \$4,500 minimum + open bar — 18 events/yr = \$81k incremental
Limited release / variant strategy	Beer-style demand + barrel-aging capacity	Quarterly bourbon barrel imperial stout — pre-sells 800 bottles at \$24 = \$19k zero-risk revenue
Festival / circuit participation	Named regional beer festivals + booth costs	Great Taste of the Midwest — \$400 booth — 8,000 attendees, captures 200 new newsletter signups
Brewery merch line	Local apparel printer + design + margin	T-shirts + hats + branded glassware — \$24 shirts cost \$7 — 800 units/yr = \$13.6k margin
Loyalty / app program	Untappd partnership + check-in rewards	"Untappd 25 check-ins → free flight" — drives repeat visits + UGC reviews
Dog-friendly family-hour positioning	County health dept rules + family demand	Sun 12-4 family-friendly — kids menu + lawn games — fills traditionally dead Sundays
Charity collab beer / pint nights	Named local nonprofits + beneficiary structure	\$1/pint to local food bank — Tue night — drives 22% volume lift + community goodwill

5. Specificity rules (L4 — what 'real named entity' means here)

- Every beer / spirit recommendation must reference a real style + ABV target
- Every distribution partner must be a real licensed distributor in this state
- Every named festival must be currently active with a verifiable URL
- Every cert / member program must be real (Brewers Association, BJCP)

6. Cost scale (L5 — per-sector currency bands)

Bar / brewery operations evaluate at hospitality + production bands:

1 — Free or under \$200	Tap takeover, event programming, partner outreach
2 — \$200 to \$2,000	Mug club setup, photo shoot, event marketing
3 — \$2,000 to \$20,000	Canning equipment, patio buildout, merch line
4 — \$20,000 to \$100,000	Brewing capacity expansion, second location buildout
5 — Over \$100,000	New brewhouse, distribution warehouse, full taproom buildout

7. Persona schema (L6)

Schema: B2C_consumer

Required persona fields:

- Name + age + drink preference
- Group composition (date / friends / family / solo)
- Frequency (weekly regular vs occasional)
- Style preference + new-style willingness
- Acquisition channel (Untappd, IG, walk-by, word-of-mouth)
- LTV (visits/yr × ASP)

8. Cycle model (L7 — replaces calendar seasons)

Bar / brewery cycles run on weekly + seasonal + style cycles:

Mon-Wed lulls	Weekday slow nights	Trivia, industry night, can release
Thu-Sat peaks	Weekend volume	Live music, food truck, full programming
Sun afternoon	Family / casual + brunch	Brunch beer, family-friendly hours, NFL Sunday
Spring (Mar-May)	Patio open + IPA season	Patio launch event, hop-forward releases
Summer (Jun-Aug)	Peak patio + tourism	Light beers + kettle sours, outdoor festival presence
Fall (Sep-Nov)	Oktoberfest + harvest beers	Seasonal release, stein-hold competition
Winter (Dec-Feb)	Stout + barrel-aged season	Imperial stout release, holiday markets

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Slow bar service during peak — universal — frame as bartender ratio + POS efficiency
- Inconsistent draft quality — universal — frame as line cleaning + glass washing protocols
- Beer freshness for distributed accounts — universal — frame as rotation discipline
- Loud / noise complaints — universal — frame as acoustic + neighbor relations
- Parking + transit access — universal — frame as ride-share partnership
- Underage attempt frequency — universal — frame as ID training + scanner

10. Report sections (L9 — section list and order)

- 1 **Establishment snapshot** — Beer styles on tap, ASP, weekly traffic, named comp set
- 2 **Top 5 unique programming + product plays** — Drawn from 14-category playbook
- 3 **Quick wins (under \$2k)** — Process + partnership wins
- 4 **Distribution + off-premise plan** — Named accounts + can / growler economics
- 5 **Customer personas (4 + lost guest)** — Regular / weekend / event-driven / tourist
- 6 **Annual + weekly cycle plan** — Style + event calendar
- 7 **Comp set comparison** — Named brewery / bar peers + their gaps
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Avg pints poured per day
- Avg ticket per visit
- Member / mug club enrollment count
- Untappd avg rating + check-in count
- Off-premise revenue % (distribution + growler)
- Event nights occupancy %

Engineering note for this sector

Sector 72 is the v3 sweet spot — engineers should treat each sub-cluster spec as a structured serialization of what `server.js` already does. The work is refactoring v3's hospitality logic into the `sector_profile` shape, not building new analysis. Estimated effort: 2-3 weeks for 1 engineer to lift-and-shift all 3 sub-cluster profiles from existing code.

CHAPTER 81 — OTHER SERVICES

Auto repair, personal care (salons / barbers / nails / spa), funeral homes, laundry, pet care non-vet, religious / civic. Three distinct profiles plus a niche.

Covers 44 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.

The v3 problem in this sector

Auto repair runs on trade-pattern logic (HVAC-style trust + emergency). Personal care (salons) runs on hospitality logic (recurring B2C appointment). Funeral homes run on once-per-decade-per-family sensitivity. Three radically different profiles inside one NAICS sector.

Sub-clusters in this sector

This sector splits into 3 sub-clusters because the business models inside diverge enough that one profile cannot serve them all. Each sub-cluster gets its own complete spec below.

Profile ID	Business Model	NAICS Codes
otherservices.auto_repair	Auto repair, body shops, oil change, tire shops, transmission specialists	B2C_appointment 811111, 811114, 811121, 811122, 811191, 811192, 811198
otherservices.personal_care	Hair salons, barbershops, nail salons, day spas, lash / brow / wax studios	B2C_appointment 812111, 812112, 812113, 812199
otherservices.funeral	Funeral homes, crematories, cemetery services	B2C_appointment 812210, 812220

81.auto_repair — Auto repair, body shops, oil change, tire shops, transmission specialists

Business model: B2C_appointment | **Profile ID:** otherservices.auto_repair

Covers NAICS: 811111, 811114, 811121, 811122, 811191, 811192, 811198

Modern formats (Part 2): EV-Specialist Repair, Mobile Mechanic, Specialty Brand Shop (BMW / Tesla / European)

1. Data sources (replaces v3 hospitality stack)

Data Source	Details	Cost	Access
Google Places + Reviews	Reviews, photos, hours	\$0.02	Yelp
BBB + state DOT licensing	License + complaint history	Free	Direct office

ASE certification database	Named technician credentials	Free	Direct
NAPA AutoCare + ACDelco directories	Named warranty programs	Free	Manufacturer
CARFAX shop network	Service history visibility	Subscription	Direct
Mitchell / AllData / Identifix	Estimating + diagnostic data	Subscription	OEM TSBs
Local fleet + dealer service	Named potential commercial accounts	Free	Direct call

2. Competitor radius

5 mi urban, 12 mi suburban, 20 mi rural. Specialty (BMW, Tesla, EV) competes 30+ mi. Mobile mechanics defined by service-area, not radius.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Diagnostic accuracy
- Repair quality / warranty
- Pricing transparency
- Communication during repair
- Wait / loaner availability
- Cleanliness of waiting area
- Honesty (don't up-sell unnecessary)
- Speed vs estimate

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Digital vehicle inspection program (DVI)	AutoVitals / Mitchell DVI tools + benchmark conversion	AutoVitals DVI photos + estimate to customer phone — average ARO from \$340 to \$480 (+41%)
Online appointment booking	Named scheduling tools (AutoLeap, Tekmetric, Shop-Ware)	AutoLeap online booking — captures 28% of appointments after-hours, fills 4 unbooked bays/wk
Loaner / shuttle program	Loaner fleet economics + repeat business lift	4-car loaner fleet (\$400/mo each) — repeat business +18%, comeback work +24%
ASE certification + technician training	ASE blue seal program + customer-trust marketing	ASE Blue Seal of Excellence — \$0 application — drives 14% premium trust + insurance / fleet contracts

NAPA AutoCare warranty membership	24mo/24k national warranty + co-op marketing	NAPA AutoCare — \$2,400/yr — captures national warranty + co-op marketing dollars
Fleet / commercial contracts	Named local fleet operators + service contract structure	Named utility fleet — 80 vehicles — \$14k/mo recurring at 65% margin = \$109k/yr contribution
Reputation management + review velocity	Podium / Birdeye + review request automation	Podium SMS review request post-repair — Google rating from 4.1 to 4.7 in 6 mo
EV / hybrid service capability	OEM + ASE EV training + state grant programs	EV-certified tech + lift adapter + diag tool — \$14k investment — captures 28% growing EV repair share
Pre-purchase inspection (PPI) niche	Used-car private market + inspection demand	\$189 PPI — 80/mo — \$15k/mo + 22% conversion to follow-up repair work
Maintenance subscription program	Subscription model in service (Christian Brothers, etc.)	\$29/mo unlimited oil changes + tire rotation — 220 subscribers = \$6,380/mo recurring
Tire program + alignment add-on	Named tire programs (BG, Goodyear, Bridgestone) + margins	Bridgestone Tireway dealer program — captures \$12k/mo tire revenue at 22% margin
Insurance / dealer service overflow	Named dealer service + insurance adjuster networks	Named dealer service overflow contract — \$14k/mo at preferred rates
Brand specialization positioning	OEM-specific tools + parts + tech training	"BMW Specialist" positioning — captures premium \$/hr from independent BMW owners
Customer waiting amenities	Per-visit experience benchmarks	WiFi + comfortable seating + complimentary coffee / car wash — review sentiment lift 12%

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named program must be real (ASE, NAPA AutoCare, ACDelco PSC, Bosch Service)
- Every named software / tool must be real (AutoVitals, Mitchell, AllData, AutoLeap, Tekmetric)
- Every named tire / parts program must be a real distributor program
- Every benchmark must come from ATD / Automotive News / NADA Service Yearbook

6. Cost scale (L5 — per-sector currency bands)

Auto repair shop spend bands are small-business + equipment scale:

1 — Under \$500	Cert application, scheduling software trial, photo shoot
2 — \$500 to \$5,000	DVI software setup, signage, Podium / Birdeye, training
3 — \$5,000 to \$50,000	Lift add, alignment machine, EV diag tool, loaner car

4 — \$50,000 to \$500,000	Bay expansion, scan tools, paint booth, warehouse build
5 — Over \$500,000	Second location, full re-equip, body shop add

7. Persona schema (L6)

Schema: B2C_consumer (with vehicle-lifecycle + service-loyalty fields)

Required persona fields:

- Name + vehicle make / model / year / mileage
- Service preference (dealer vs independent vs DIY)
- Trust trigger (transparency, photos, named tech)
- Frequency (DIY occasional vs every-service customer)
- Acquisition channel (Google, friend, dealer overflow, fleet)
- LTV (services/yr × ARO × loyalty years)

8. Cycle model (L7 — replaces calendar seasons)

Auto repair runs on weather + service-interval + tax cycles:

Pre-winter (Sep-Nov)	Tire + battery + heater + winterization	Targeted email + tire promo + battery test campaign
Pre-summer (Mar-May)	AC + brakes + tire rotation	AC service promo + spring inspection campaign
Tax-refund (Feb-Apr)	Customer cash availability for big repairs	Promote deferred-repair completion
Back-to-school (Aug)	Family vehicle prep	Multi-vehicle inspection package
Service-interval comebacks	5k mi, 30k mi, 60k mi, 100k mi	Automated reminders by mileage
Weekly: peak demand	Mon AM + Fri PM rush	Schedule discipline + same-day-completion guarantee
Annual: ASE cert renewal	Every-2-yr cert renewal	Tech retention + cert maintenance budget

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Surprise pricing / scope creep — universal — frame as DVI + photo estimate
- Slow communication during repair — universal — frame as text-message updates
- Comeback / warranty issues — universal — frame as written warranty + national NAPA program
- No-shows + scheduling chaos — universal — frame as deposit + reminder
- Loaner / shuttle gap — universal — frame as fleet ROI

- Up-sell pressure — universal — frame as menu pricing + opt-in mode
- Female customer comfort — universal — frame as inclusive + transparent shop culture

10. Report sections (L9 — section list and order)

- 1 **Shop snapshot** — Bays, ARO, monthly RO count, named comp set
- 2 **Top 5 trust + revenue moves** — Drawn from 14-category playbook
- 3 **Service department deep-dive** — DVI + scheduling + comm + reputation tactics
- 4 **Customer personas (4 + lost customer)** — DIY-supplement / regular / fleet / female
- 5 **Fleet + commercial pipeline** — Named prospects + contract structure
- 6 **Reputation + comm strategy** — Named platforms + tactic-by-platform
- 7 **Annual cycle plan** — Weather + service-interval calendar
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Avg repair order (ARO) \$
- Comeback rate (target < 2%)
- Customer retention 12-mo
- Bay utilization %
- Online booking % of appointments
- Avg Google rating + review velocity

81.personal_care — Hair salons, barbershops, nail salons, day spas, lash / brow / wax studios

Business model: B2C_appointment | **Profile ID:** otherservices.personal_care

Covers NAICS: 812111, 812112, 812113, 812199

Modern formats (Part 2): Lash / Brow Studio, Wax Studio, Spray-Tan, Med-Spa (basic)

1. Data sources (replaces v3 hospitality stack)

Google Places + Reviews	Reviews, photos, hours	\$0.02	Yelp	
Yelp Fusion API	Reviews + categories	\$0.01	Manual	
Booksy / GlossGenius / Vagaro	Booking demand + ASP benchmarks	Subscription public	Direct	
Salon Today / Modern Salon	Industry trend + benchmarks	Subscription	Public	
Instagram + TikTok	Stylist personal-brand following	Free	Manual	
Foursquare	Foot traffic + dwell	Subscription	Door count	
Local cosmetology school	Apprentice / apprentice talent pipeline	Free	Direct call	

2. Competitor radius

2 km urban, 6 km suburban, 15 km rural. Specialty (color, curls, lashes) draws 20+ km. High-end / med-spa hybrid 25+ km.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Stylist skill + specialty match
- Cleanliness + sanitation
- On-time appointment
- Pricing transparency
- Salon vibe + atmosphere
- Booking ease
- Product retail + recommendation
- Consultation quality

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Stylist personal-brand build	IG + TikTok creator economics in beauty	Each stylist runs own IG — 4× book-rate from social referral — adds 2 fills/wk × \$128 = \$1,024/wk
Online booking + waitlist	Booksy / GlossGenius / Vagaro feature comparison	Online booking 24/7 — captures 28% of bookings after-hours, fills 6 chairs/wk
Retail / product attach	Pro-line product (Aveda, Olaplex, R+Co) + margin	Olaplex retail — \$32/bottle — 14% attach × 80 clients/wk = \$358/wk × 52 = \$18.6k margin
Membership / VIP pass program	Successful peer membership programs	\$95/mo VIP — unlimited blowouts + 10% off services — 80 members = \$91k/yr predictable
Pre-book / standing appointment	Industry benchmark pre-book rate	"Book your next 6-week visit before you leave" — pre-book rate from 22% to 64% — fills 4-week gaps
Specialty service expansion	Adjacent service margin + training cost	Add lash + brow tinting — \$48 service × 60% margin × 12/wk = \$345/wk on existing chair time
Bridal / wedding party packages	Named local wedding venues + planner partnerships	Bridal package — \$480 bride + 6 BMs at \$128 = \$1,248 — 14 weddings × 6mo busy = \$17.5k
Booth-rent vs commission analysis	Industry comp + state regulations	Hybrid: 50% commission + booth-rent option — retains best stylists + reduces W2 burden
Color / chemical specialty positioning	Color brand training + Olaplex / Wella / Redken	Master Colorist designation — \$360 hair appointment vs \$180 standard — 4 specialty bookings/wk = \$720/wk uplift
Subscription drip / loyalty	Software-integrated loyalty + email	Birthday + anniversary email with \$20 reward — 22% redemption rate
Influencer / press partnership	Named local + regional beauty press	Featured stylist — named local IG creator (28k) — 6-mo partnership = 60+ new clients
Apprentice / assistant pipeline	Named local cosmetology school	Hire 2 apprentices/yr from named cosmetology school — covers shampoo + assist for senior stylists
Pricing tier + level system	Stylist-tier pricing across major chains	Junior / Senior / Master tier system — junior \$48 vs Master \$148 — captures both ends
On-trend launches (gloss, balayage, lived-in color)	Trend tracking + training cycle	Launch lived-in color treatment — \$180 service × 4/wk = \$720/wk new revenue stream

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named product line must be real (Aveda, Redken, Oribe, Olaplex, R+Co, Schwarzkopf)
- Every named software must be real (Booksy, GlossGenius, Vagaro, Mindbody, Square Appointments)

- Every named cosmetology school must be a verifiable local school
- Every benchmark must come from PBA / Salon Today / Modern Salon

6. Cost scale (L5 — per-sector currency bands)

Personal care spend bands are small-business hospitality scale:

1 — Under \$200	Software trial, partnership outreach, photo shoot
2 — \$200 to \$2,000	Booking software setup, pro-line retail starter, signage
3 — \$2,000 to \$20,000	Salon refresh, color line addition, marketing campaign
4 — \$20,000 to \$100,000	Full salon renovation, second chair / station, suite buildout
5 — Over \$100,000	Second location, full med-spa add, new buildout

7. Persona schema (L6)

Schema: B2C_consumer

Required persona fields:

- Name + age + service profile (color, cut, treatment)
- Frequency (4-wk vs 8-wk vs occasional)
- Stylist loyalty (chair-loyal vs salon-loyal vs neither)
- Acquisition channel (IG, walk-in, friend, search)
- Retail attach behavior
- LTV (visits/yr × ASP × loyalty years)

8. Cycle model (L7 — replaces calendar seasons)

Personal care runs on holiday + wedding + back-to-school cycles:

Holiday season (Nov-Dec)	Party + family-photo demand	Holiday packages, gift card push, blowout / lash bookings	
Wedding season (May-Sep)	Bridal + wedding-party demand	Bridal packages, trial appointments, planner partnerships	
Back-to-school (Aug)	Family + kids haircuts	Family bundle pricing, school-day promos	
Pre-summer (Apr-May)	Beach-prep + lighter color	Highlights + balayage promo	
Weekly: peak times	Thu-Sat heavy, Mon-Tue light	Mon-Tue promo + fill-in pricing	
Pre-event (prom, gala, milestone)	Local school / community calendar	Targeted email by ZIP + age	
Annual: stylist training + reviews	Continuing education + retention	Pay raises tied to skill + revenue contribution	

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Late appointments / overbooking — universal — frame as scheduling discipline
- Inconsistent quality across stylists — universal — frame as training + leveling system
- Hidden fees / surprise total — universal — frame as upfront pricing + consultation
- Stylist turnover (clients leave with stylist) — universal — frame as commission / culture / non-compete
- Cleanliness / sanitation perception — universal — frame as visible cleaning protocols
- Booking system confusion — universal — frame as online booking + reminder
- Retail pressure complaints — universal — frame as recommendation vs sale framing

10. Report sections (L9 — section list and order)

- 1 **Salon snapshot** — Chairs, monthly tickets, ASP, retention, named comp set
- 2 **Top 5 growth + retention moves** — Drawn from 14-category playbook
- 3 **Stylist build + retention plan** — Tier + commission + personal-brand support
- 4 **Customer personas (4 + lost client)** — New / regular / VIP / lapsed
- 5 **Service menu + retail audit** — Pricing tier + retail attach plan
- 6 **Annual cycle plan** — 7-window calendar tied to local events
- 7 **Comp set comparison** — Named salons + their gaps
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Avg ticket per appointment
- Pre-book rate (target > 60%)
- Retail attach % (target > 12%)
- Stylist retention (12-mo)
- New-client retention (90-day return rate, target > 50%)
- Chair utilization %

81.funeral — Funeral homes, crematories, cemetery services

Business model: B2C_appointment | **Profile ID:** otherservices.funeral

Covers NAICS: 812210, 812220

1. Data sources (replaces v3 hospitality stack)

Google Reviews	Places +	Reviews + photos	\$0.02	Yelp
NFDA directory	member	Industry benchmarks + members	Member-only	State assoc.
State funeral licensing	board	License + complaint history	Free	Direct office
Cremation Association of America	of N.	Cremation rate by state + benchmarks	Member-only	Public summary
Local clergy + church directory		Named local faith partners	Free	Direct call
Local hospice + senior community		Named pre-need referral partners	Free	Direct call

2. Competitor radius

15 mi urban, 25 mi suburban, 50 mi rural. Cremation-only providers operate wider radius (50+ mi). Faith-specific homes draw within community geography regardless of distance.

3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Compassion / empathy of staff
- Pricing transparency
- Service planning flexibility
- Faith / cultural accommodation
- Facility cleanliness + dignity
- Communication during planning
- Aftercare follow-up

4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **12** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

Pre-need plan sales program	Named insurer / trust partners + commission structure	Forethought / Homesteaders pre-need — 80 plans/yr × \$4,800 avg × 18% commission = \$69k/yr
Cremation-simplification offering	Direct-cremation pricing benchmark + capacity	\$1,295 direct cremation package vs \$7k traditional — captures 22% of market shifting toward cremation
Memorial celebration-of-life facility	Family demand for non-traditional services	Reception hall add — \$1,200 facility fee × 60 events/yr = \$72k incremental
Faith / multilingual specialization	Named faith communities + community size	Bilingual Spanish service capacity — captures growing Hispanic Catholic community within 25 mi
Live-stream service capability	OneRoom / Tukios / streaming setup	Tukios live-stream service — \$200 add-on — 60% of services elect — \$14k/yr at 100% margin
Online obituary + memorial platform	Tribute / Tukios / Frontrunner	Online memorial pages — drives 14% pre-need leads via family-network exposure
Veteran honor program	VA benefits + named veteran organizations	VFW / American Legion partnership — captures veteran burial business + builds community trust
Pet cremation / aquamation	Pet death-care market + capacity utilization	Pet cremation add-on — \$295 avg — 200/yr = \$59k revenue at low marginal cost on existing equipment
Aftercare grief support program	Named local grief counselors + group programs	Quarterly grief support group + bereaved family quarterly check-in calls — drives 8% repeat family business
Hospice senior-community partnerships	Named hospice agencies + senior living	Named hospice partner — \$0 marketing — captures 4-6 family referrals/mo
Green burial / aquamation offering	GBC certification + state aquamation rules	Green Burial Council Tier-2 cert — \$0 application — captures eco-conscious 5% market segment
Pricing transparency + GPL online	FTC Funeral Rule + online disclosure	Posted GPL on website — meets FTC Funeral Rule + builds trust + Google ranking signal

5. Specificity rules (L4 — what 'real named entity' means here)

- Every named insurer / pre-need partner must be real (Forethought, Homesteaders, Funeral Directors Life)
- Every named platform must be real (Tukios, OneRoom, Frontrunner, Tribute Center)
- Every named association must be real (NFDA, ICCFA, Cremation Association of NA)
- Every named hospice / senior community must be a verifiable local partner

6. Cost scale (L5 — per-sector currency bands)

Funeral home spend bands are small-business + facility scale:

1 — Under \$500	Pre-need agreement, partner outreach, GPL website update
2 — \$500 to \$5,000	Live-stream equipment, software setup, signage
3 — \$5,000 to \$50,000	Reception hall finish, cremation retort partnership
4 — \$50,000 to \$500,000	Cremation retort install, vehicle add, facility refresh
5 — Over \$500,000	Full facility build, second location, cemetery acquisition

7. Persona schema (L6)

Schema: B2C_consumer (with grieving + decision-committee fields)

Required persona fields:

- Name + relationship to decedent + age
- Decision authority within family
- Faith / cultural traditions to honor
- Pre-need vs at-need scenario
- Budget + financing willingness
- Aftercare needs

8. Cycle model (L7 — replaces calendar seasons)

Funeral home cycles are weather + season + community-event driven (not sales-cycle):

Winter (Dec-Mar)	Highest mortality season	Maximum staff coverage + facility readiness
Spring (Apr-May)	Average mortality + weather improves	Outdoor service capacity + grounds prep
Summer (Jun-Aug)	Lower mortality + travel	Pre-need sales focus + community engagement
Fall (Sep-Nov)	Mortality rising + weather concerns	Pre-need + community partnerships ramp
Pre-need: ongoing	Year-round community engagement	Speakers at senior centers + church groups
Annual: FTC compliance + license	GPL update + state license renewal	Compliance walk-through + GPL printing
Memorial / Veterans Day / community days	Community event participation	Sponsor + present at named local events

9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Pricing surprise / Funeral Rule violations — universal — frame as transparent GPL
- Family-decision conflict / committee dynamics — universal — frame as decision facilitation
- Cremation rate growth disrupting traditional model — universal — frame as offering both pathways
- Aftercare / grief support gap — universal — frame as community partnership
- Online presence weak — universal — frame as obit + tribute platform
- Faith / cultural accommodation gaps — universal — frame as staff training + bilingual
- Pre-need sale ethics — universal — frame as no-pressure consultation

10. Report sections (L9 — section list and order)

- 1 **Funeral home snapshot** — Annual services, cremation %, average rev/case, named comp set
- 2 **Top 5 strategic + community moves** — Drawn from 12-category playbook
- 3 **Pre-need program build / refresh** — Named provider + community outreach plan
- 4 **Customer personas (4 + lost family)** — Pre-need / at-need traditional / cremation / faith-specific
- 5 **Community + referral partnership map** — Named hospice / senior / faith partners
- 6 **Service modality + facility audit** — Cremation + green + live-stream + reception capacity
- 7 **Pricing + FTC compliance check** — GPL transparency + website disclosure
- 8 **Known industry challenges** — Collapsed drawer

Key performance indicators worth tracking

- Annual case count
- Cremation % of cases
- Average revenue per case
- Pre-need sales count + value (annual)
- Aftercare engagement rate
- Community partnership count (active)

Out-of-scope NAICS codes inside this sector

These codes belong to this sector but BizRadar refuses to analyze them — either because they are not real businesses, are regulated monopolies BizRadar cannot help, or do not have customer-facing operations to optimize.

81391x	Religious / civic / advocacy organizations — nonprofit, not customer-facing business in BizRadar sense
81411x	Private households — not businesses

Engineering note for this sector

Sector 81 splits into 3 wildly different profiles. Build otherservices.personal_care first (Phase 1, fits v3 best), then otherservices.auto_repair (Phase 2, similar to construction trades), then otherservices.funeral (Phase 5, sensitive niche). Pet care is similar enough to personal care that we can fold it under that profile with minor tweaks.